Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:28:22

Measurement Details:

- **Sample:** Muestra10
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

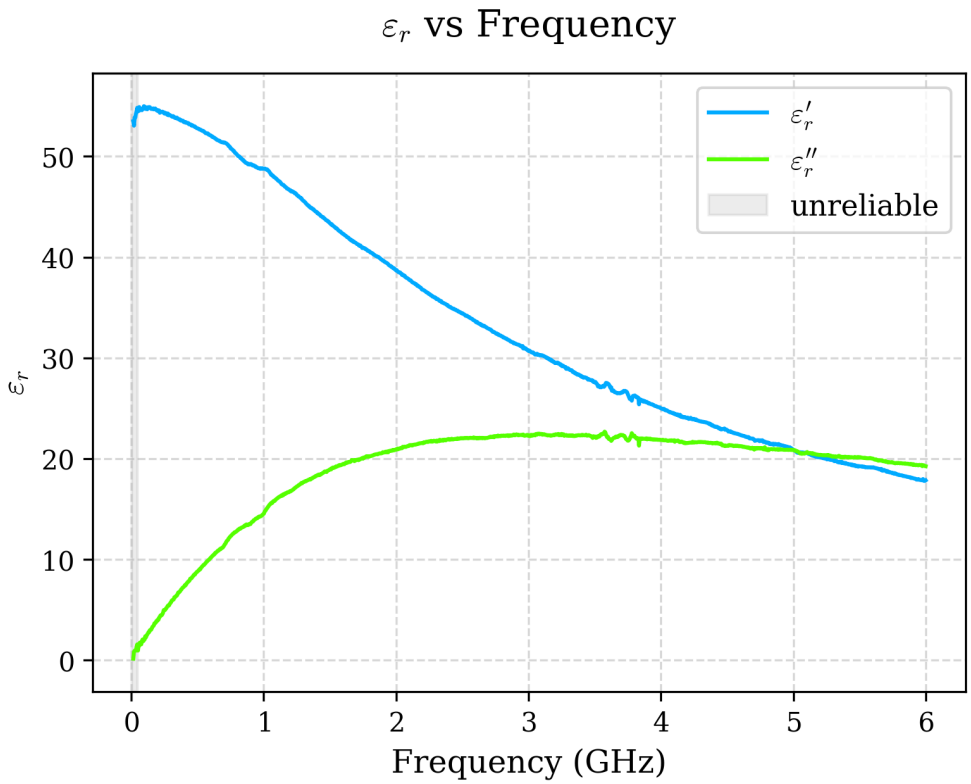
Contents

1 Measurement Results **2**
1.1 Complex Permittivity $\varepsilon_r(f)$ 2
1.2 S_{11} — Smith Chart 29

2 Measurement Quality **56**

1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	53.4886	0.1481	0.0028
0.020	53.0525	0.8304	0.0157
0.025	53.8649	0.9328	0.0173
0.030	53.8671	0.9588	0.0178
0.035	54.2387	0.9520	0.0176
0.040	54.4826	1.3390	0.0246
0.045	54.8277	1.5638	0.0285
0.050	54.4653	0.9490	0.0174
0.055	54.5201	1.4061	0.0258
0.060	54.9029	1.4392	0.0262
0.065	54.5850	1.7441	0.0320
0.070	54.6893	1.6136	0.0295
0.075	54.7349	1.5650	0.0286
0.080	54.5903	1.8885	0.0346
0.085	54.7337	2.0486	0.0374

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	54.7548	1.8779	0.0343
0.095	54.9867	1.9691	0.0358
0.100	54.9549	2.0429	0.0372
0.105	54.7151	2.3610	0.0432
0.110	54.8486	2.2527	0.0411
0.115	54.7830	2.4268	0.0443
0.120	54.6825	2.5451	0.0465
0.125	54.8831	2.6365	0.0480
0.130	54.8613	2.7784	0.0506
0.135	54.8976	2.7854	0.0507
0.140	54.8902	2.9751	0.0542
0.145	54.7704	2.9890	0.0546
0.150	54.7142	3.0224	0.0552
0.155	54.7157	3.1646	0.0578
0.160	54.7182	3.3138	0.0606
0.165	54.7677	3.3698	0.0615
0.170	54.6761	3.4554	0.0632
0.175	54.7022	3.6600	0.0669
0.180	54.7611	3.6705	0.0670
0.185	54.6983	3.7296	0.0682
0.190	54.5895	3.8910	0.0713
0.195	54.5479	3.8774	0.0711
0.200	54.5104	4.0210	0.0738
0.205	54.5882	4.0061	0.0734
0.210	54.5460	4.2196	0.0774
0.215	54.3372	4.2872	0.0789
0.220	54.4446	4.4358	0.0815
0.225	54.4333	4.4321	0.0814
0.230	54.4085	4.4643	0.0821
0.235	54.4474	4.5692	0.0839
0.240	54.4030	4.7206	0.0868
0.245	54.3627	4.9635	0.0913
0.250	54.2541	4.9917	0.0920
0.255	54.3326	4.9785	0.0916
0.260	54.3029	5.0708	0.0934
0.265	54.2283	5.1703	0.0953
0.270	54.1528	5.2699	0.0973
0.275	54.2653	5.3822	0.0992
0.280	54.0878	5.4446	0.1007
0.285	54.1487	5.4322	0.1003
0.290	54.1086	5.5659	0.1029
0.295	54.0662	5.6771	0.1050
0.300	54.0352	5.7374	0.1062
0.305	53.9736	5.9071	0.1094
0.310	53.9373	5.9560	0.1104
0.315	54.0053	5.9794	0.1107

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	53.9634	6.0282	0.1117
0.325	53.8720	6.2399	0.1158
0.330	53.8254	6.2445	0.1160
0.335	53.8253	6.3137	0.1173
0.340	53.7842	6.3827	0.1187
0.345	53.8466	6.4219	0.1193
0.350	53.7529	6.5885	0.1226
0.355	53.7053	6.7151	0.1250
0.360	53.6908	6.7135	0.1250
0.365	53.6939	6.7948	0.1265
0.370	53.5821	6.8628	0.1281
0.375	53.6005	6.9959	0.1305
0.380	53.5624	7.0453	0.1315
0.385	53.5538	7.1448	0.1334
0.390	53.5444	7.2204	0.1348
0.395	53.4276	7.2566	0.1358
0.400	53.3941	7.3196	0.1371
0.405	53.4272	7.4533	0.1395
0.410	53.4130	7.4271	0.1391
0.415	53.3341	7.5458	0.1415
0.420	53.3357	7.6427	0.1433
0.425	53.3227	7.7210	0.1448
0.430	53.3197	7.8064	0.1464
0.435	53.1984	7.8328	0.1472
0.440	53.2038	7.9174	0.1488
0.445	53.1792	8.0843	0.1520
0.450	53.1622	8.1146	0.1526
0.455	53.0618	8.1268	0.1532
0.460	53.1039	8.2439	0.1552
0.465	53.0458	8.3098	0.1567
0.470	53.0253	8.4185	0.1588
0.475	52.9659	8.4925	0.1603
0.480	52.9750	8.4884	0.1602
0.485	52.8950	8.6065	0.1627
0.490	52.8553	8.6763	0.1642
0.495	52.8036	8.7582	0.1659
0.500	52.7860	8.7926	0.1666
0.505	52.7681	8.8625	0.1680
0.510	52.7229	8.9076	0.1690
0.515	52.6763	9.0211	0.1713
0.520	52.6742	9.0110	0.1711
0.525	52.6315	9.1219	0.1733
0.530	52.5958	9.2286	0.1755
0.535	52.5466	9.2844	0.1767
0.540	52.5469	9.3667	0.1783
0.545	52.5225	9.4459	0.1798

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	52.4447	9.4633	0.1804
0.555	52.4043	9.5430	0.1821
0.560	52.3208	9.6305	0.1841
0.565	52.2984	9.6915	0.1853
0.570	52.2775	9.7756	0.1870
0.575	52.2997	9.8293	0.1879
0.580	52.2578	9.9109	0.1897
0.585	52.1623	9.9520	0.1908
0.590	52.1308	10.0064	0.1919
0.595	52.1052	10.1070	0.1940
0.600	52.0880	10.1705	0.1953
0.605	52.0262	10.2259	0.1966
0.610	51.9513	10.3121	0.1985
0.615	51.9514	10.3780	0.1998
0.620	51.9004	10.4131	0.2006
0.625	51.8537	10.5037	0.2026
0.630	51.8310	10.5293	0.2031
0.635	51.7687	10.6614	0.2059
0.640	51.7317	10.6898	0.2066
0.645	51.6499	10.7328	0.2078
0.650	51.6246	10.8135	0.2095
0.655	51.6076	10.8773	0.2108
0.660	51.5715	10.8808	0.2110
0.665	51.5602	10.9502	0.2124
0.670	51.4893	11.0409	0.2144
0.675	51.4651	11.0189	0.2141
0.680	51.4234	11.1085	0.2160
0.685	51.4043	11.1425	0.2168
0.690	51.4165	11.1627	0.2171
0.695	51.3780	11.2525	0.2190
0.700	51.3618	11.3534	0.2210
0.705	51.3873	11.4754	0.2233
0.710	51.3455	11.5474	0.2249
0.715	51.3030	11.6729	0.2275
0.720	51.2733	11.8083	0.2303
0.725	51.2087	11.8741	0.2319
0.730	51.1545	11.9328	0.2333
0.735	51.0312	12.0697	0.2365
0.740	51.0001	12.1523	0.2383
0.745	50.9430	12.2270	0.2400
0.750	50.7977	12.3102	0.2423
0.755	50.7596	12.3579	0.2435
0.760	50.7157	12.4423	0.2453
0.765	50.6508	12.5108	0.2470
0.770	50.5676	12.5759	0.2487
0.775	50.5433	12.5968	0.2492

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	50.4273	12.6915	0.2517
0.785	50.3360	12.7137	0.2526
0.790	50.3003	12.7367	0.2532
0.795	50.2372	12.7974	0.2547
0.800	50.1430	12.8443	0.2562
0.805	50.0620	12.9373	0.2584
0.810	50.0264	12.9515	0.2589
0.815	50.0086	13.0284	0.2605
0.820	49.9305	13.0776	0.2619
0.825	49.8367	13.0564	0.2620
0.830	49.8314	13.1265	0.2634
0.835	49.7367	13.1747	0.2649
0.840	49.6608	13.1882	0.2656
0.845	49.6522	13.2204	0.2663
0.850	49.5338	13.2525	0.2675
0.855	49.5087	13.3526	0.2697
0.860	49.4094	13.3207	0.2696
0.865	49.3637	13.4330	0.2721
0.870	49.3477	13.4662	0.2729
0.875	49.3284	13.3997	0.2716
0.880	49.2340	13.3971	0.2721
0.885	49.2066	13.4448	0.2732
0.890	49.2412	13.5108	0.2744
0.895	49.1908	13.4965	0.2744
0.900	49.2491	13.5043	0.2742
0.905	49.2411	13.6275	0.2768
0.910	49.1885	13.7037	0.2786
0.915	49.1115	13.7429	0.2798
0.920	49.0977	13.7654	0.2804
0.925	49.1145	13.8947	0.2829
0.930	49.0757	13.9235	0.2837
0.935	48.9908	13.9749	0.2853
0.940	49.0092	14.0473	0.2866
0.945	48.9192	14.0760	0.2877
0.950	48.8993	14.0696	0.2877
0.955	48.8699	14.1654	0.2899
0.960	48.8196	14.1809	0.2905
0.965	48.8290	14.2534	0.2919
0.970	48.8288	14.2555	0.2919
0.975	48.7708	14.2585	0.2924
0.980	48.8143	14.3073	0.2931
0.985	48.8235	14.3481	0.2939
0.990	48.7730	14.4193	0.2956
0.995	48.7952	14.4672	0.2965
1.000	48.7870	14.5162	0.2975
1.005	48.7819	14.6715	0.3008

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	48.7919	14.8052	0.3034
1.015	48.7230	14.8674	0.3051
1.020	48.7023	14.9891	0.3078
1.025	48.7243	15.0724	0.3093
1.030	48.6279	15.1660	0.3119
1.035	48.5996	15.2553	0.3139
1.040	48.5079	15.3600	0.3166
1.045	48.4045	15.4278	0.3187
1.050	48.3318	15.5206	0.3211
1.055	48.1990	15.5301	0.3222
1.060	48.1885	15.6434	0.3246
1.065	48.1026	15.6056	0.3244
1.070	48.0831	15.7196	0.3269
1.075	47.9750	15.8019	0.3294
1.080	47.9462	15.7928	0.3294
1.085	47.8906	15.8263	0.3305
1.090	47.8608	15.9210	0.3327
1.095	47.7704	15.9917	0.3348
1.100	47.7319	16.0381	0.3360
1.105	47.6541	16.0712	0.3372
1.110	47.5815	16.1488	0.3394
1.115	47.5485	16.1958	0.3406
1.120	47.5120	16.2176	0.3413
1.125	47.4483	16.2435	0.3423
1.130	47.3552	16.3179	0.3446
1.135	47.2527	16.3477	0.3460
1.140	47.2166	16.3169	0.3456
1.145	47.1742	16.4418	0.3485
1.150	47.1663	16.3912	0.3475
1.155	47.0837	16.4940	0.3503
1.160	46.9961	16.4785	0.3506
1.165	46.9716	16.5643	0.3526
1.170	46.9142	16.5663	0.3531
1.175	46.8113	16.6016	0.3546
1.180	46.8056	16.6702	0.3562
1.185	46.7700	16.6586	0.3562
1.190	46.6979	16.6687	0.3569
1.195	46.6719	16.7315	0.3585
1.200	46.6594	16.7803	0.3596
1.205	46.5387	16.7625	0.3602
1.210	46.5746	16.8280	0.3613
1.215	46.4977	16.8727	0.3629
1.220	46.5116	16.9019	0.3634
1.225	46.4670	16.9580	0.3649
1.230	46.4371	17.0399	0.3669
1.235	46.3292	17.0703	0.3685

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	46.3495	17.1347	0.3697
1.245	46.2913	17.1859	0.3713
1.250	46.1758	17.2338	0.3732
1.255	46.1164	17.2704	0.3745
1.260	46.0931	17.3169	0.3757
1.265	46.0038	17.3868	0.3779
1.270	45.9712	17.4079	0.3787
1.275	45.9291	17.4559	0.3801
1.280	45.8489	17.5163	0.3820
1.285	45.8478	17.5915	0.3837
1.290	45.7130	17.5503	0.3839
1.295	45.6406	17.6215	0.3861
1.300	45.5994	17.6749	0.3876
1.305	45.5434	17.6882	0.3884
1.310	45.4767	17.7348	0.3900
1.315	45.4173	17.7684	0.3912
1.320	45.3194	17.7605	0.3919
1.325	45.2670	17.8126	0.3935
1.330	45.1779	17.8432	0.3950
1.335	45.1325	17.8706	0.3960
1.340	45.1170	17.9072	0.3969
1.345	44.9927	17.9167	0.3982
1.350	44.9598	17.9836	0.4000
1.355	44.8911	17.9373	0.3996
1.360	44.9033	18.0124	0.4011
1.365	44.8146	18.0651	0.4031
1.370	44.7773	18.0749	0.4037
1.375	44.7721	18.1347	0.4050
1.380	44.6603	18.1612	0.4067
1.385	44.6173	18.2168	0.4083
1.390	44.5508	18.2243	0.4091
1.395	44.5097	18.2987	0.4111
1.400	44.4952	18.3050	0.4114
1.405	44.3650	18.3504	0.4136
1.410	44.3694	18.3936	0.4146
1.415	44.2763	18.3500	0.4144
1.420	44.2603	18.4194	0.4162
1.425	44.2056	18.3863	0.4159
1.430	44.0977	18.4618	0.4187
1.435	44.0655	18.4686	0.4191
1.440	44.0122	18.5249	0.4209
1.445	43.9640	18.5604	0.4222
1.450	43.9608	18.6047	0.4232
1.455	43.8664	18.6586	0.4254
1.460	43.8285	18.7229	0.4272
1.465	43.7586	18.7298	0.4280

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	43.6877	18.7211	0.4285
1.475	43.6486	18.6741	0.4278
1.480	43.6316	18.7879	0.4306
1.485	43.5498	18.8101	0.4319
1.490	43.5311	18.8472	0.4330
1.495	43.4405	18.8676	0.4343
1.500	43.3736	18.9346	0.4365
1.505	43.3191	18.9699	0.4379
1.510	43.2325	18.9374	0.4380
1.515	43.2636	19.0677	0.4407
1.520	43.1490	19.0733	0.4420
1.525	43.1049	19.1219	0.4436
1.530	43.0630	19.0831	0.4431
1.535	43.0132	19.1190	0.4445
1.540	42.9355	19.1425	0.4458
1.545	42.8704	19.2280	0.4485
1.550	42.8764	19.2279	0.4485
1.555	42.8030	19.2212	0.4491
1.560	42.7234	19.2709	0.4511
1.565	42.6580	19.3240	0.4530
1.570	42.6040	19.3053	0.4531
1.575	42.5650	19.3711	0.4551
1.580	42.5441	19.3526	0.4549
1.585	42.4537	19.3579	0.4560
1.590	42.3937	19.4084	0.4578
1.595	42.3573	19.4315	0.4588
1.600	42.2943	19.4749	0.4605
1.605	42.2743	19.4884	0.4610
1.610	42.2137	19.4498	0.4607
1.615	42.1515	19.5385	0.4635
1.620	42.1190	19.6015	0.4654
1.625	42.0595	19.6299	0.4667
1.630	41.9850	19.5784	0.4663
1.635	41.9589	19.6430	0.4681
1.640	41.9366	19.6672	0.4690
1.645	41.8516	19.6821	0.4703
1.650	41.7726	19.6606	0.4707
1.655	41.7544	19.6839	0.4714
1.660	41.6617	19.7589	0.4743
1.665	41.6906	19.8619	0.4764
1.670	41.6043	19.8326	0.4767
1.675	41.5325	19.8073	0.4769
1.680	41.5620	19.8442	0.4775
1.685	41.4866	19.7497	0.4760
1.690	41.4548	19.8579	0.4790
1.695	41.4213	19.8366	0.4789

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	41.3602	19.8620	0.4802
1.705	41.3041	19.8950	0.4817
1.710	41.2765	19.8894	0.4819
1.715	41.2619	19.8483	0.4810
1.720	41.1607	19.9282	0.4842
1.725	41.1149	19.9045	0.4841
1.730	41.1610	19.9676	0.4851
1.735	41.0612	19.9866	0.4868
1.740	41.0002	20.0244	0.4884
1.745	40.9003	20.0629	0.4905
1.750	40.9509	19.9956	0.4883
1.755	40.9201	20.1087	0.4914
1.760	40.8761	20.0932	0.4916
1.765	40.8677	20.1394	0.4928
1.770	40.7867	20.1617	0.4943
1.775	40.7751	20.2401	0.4964
1.780	40.7192	20.1778	0.4955
1.785	40.6367	20.2256	0.4977
1.790	40.6203	20.1966	0.4972
1.795	40.6063	20.2334	0.4983
1.800	40.5564	20.2138	0.4984
1.805	40.4698	20.2503	0.5004
1.810	40.4548	20.2583	0.5008
1.815	40.4202	20.2790	0.5017
1.820	40.3837	20.3027	0.5027
1.825	40.3526	20.3299	0.5038
1.830	40.2539	20.3520	0.5056
1.835	40.2278	20.3952	0.5070
1.840	40.1835	20.4057	0.5078
1.845	40.1414	20.4243	0.5088
1.850	40.1088	20.4368	0.5095
1.855	40.0890	20.4413	0.5099
1.860	40.0322	20.4258	0.5102
1.865	40.0330	20.4732	0.5114
1.870	39.9903	20.4939	0.5125
1.875	39.9196	20.5205	0.5140
1.880	39.8791	20.5206	0.5146
1.885	39.7933	20.5199	0.5157
1.890	39.7671	20.5334	0.5163
1.895	39.7119	20.5630	0.5178
1.900	39.6528	20.6410	0.5205
1.905	39.6430	20.6104	0.5199
1.910	39.5862	20.6137	0.5207
1.915	39.5156	20.6569	0.5228
1.920	39.4862	20.6448	0.5228
1.925	39.4597	20.7015	0.5246

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	39.4126	20.6441	0.5238
1.935	39.3780	20.7073	0.5259
1.940	39.2978	20.6299	0.5250
1.945	39.2212	20.7504	0.5291
1.950	39.1736	20.7530	0.5298
1.955	39.1379	20.7826	0.5310
1.960	39.1256	20.7680	0.5308
1.965	39.0247	20.7908	0.5328
1.970	39.0059	20.7905	0.5330
1.975	38.9567	20.8698	0.5357
1.980	38.8861	20.8189	0.5354
1.985	38.8923	20.8290	0.5356
1.990	38.8249	20.8748	0.5377
1.995	38.7716	20.8961	0.5390
2.000	38.6756	20.9266	0.5411
2.005	38.6996	20.9056	0.5402
2.010	38.6116	21.0032	0.5440
2.015	38.5515	20.9276	0.5428
2.020	38.4970	20.9605	0.5445
2.025	38.4868	21.0127	0.5460
2.030	38.4838	20.9973	0.5456
2.035	38.3979	21.0530	0.5483
2.040	38.3639	21.0632	0.5490
2.045	38.2324	21.0371	0.5502
2.050	38.2538	21.1757	0.5536
2.055	38.2286	21.1559	0.5534
2.060	38.1295	21.1677	0.5552
2.065	38.1533	21.1907	0.5554
2.070	38.0447	21.1604	0.5562
2.075	37.9678	21.1701	0.5576
2.080	37.9473	21.2088	0.5589
2.085	37.9128	21.2345	0.5601
2.090	37.8703	21.2473	0.5611
2.095	37.8202	21.2535	0.5620
2.100	37.7404	21.3345	0.5653
2.105	37.7130	21.3142	0.5652
2.110	37.7020	21.3472	0.5662
2.115	37.6148	21.3339	0.5672
2.120	37.5479	21.3028	0.5673
2.125	37.4841	21.3600	0.5698
2.130	37.4624	21.3554	0.5701
2.135	37.4333	21.3889	0.5714
2.140	37.4019	21.3948	0.5720
2.145	37.3206	21.3839	0.5730
2.150	37.3303	21.4079	0.5735
2.155	37.2424	21.4214	0.5752

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	37.1564	21.4410	0.5770
2.165	37.1742	21.4757	0.5777
2.170	37.0895	21.4747	0.5790
2.175	37.0645	21.5187	0.5806
2.180	37.0063	21.5217	0.5816
2.185	36.9801	21.5331	0.5823
2.190	36.9103	21.5427	0.5836
2.195	36.7897	21.5457	0.5856
2.200	36.7696	21.5683	0.5866
2.205	36.7501	21.5319	0.5859
2.210	36.7183	21.6276	0.5890
2.215	36.6635	21.5939	0.5890
2.220	36.6264	21.6205	0.5903
2.225	36.5676	21.6642	0.5924
2.230	36.5028	21.6743	0.5938
2.235	36.4920	21.6759	0.5940
2.240	36.4614	21.6511	0.5938
2.245	36.3689	21.7132	0.5970
2.250	36.3450	21.7003	0.5971
2.255	36.3228	21.6996	0.5974
2.260	36.2330	21.7415	0.6000
2.265	36.2062	21.7667	0.6012
2.270	36.1830	21.6876	0.5994
2.275	36.0892	21.7150	0.6017
2.280	36.0753	21.7091	0.6018
2.285	36.0488	21.7790	0.6042
2.290	35.9887	21.7639	0.6047
2.295	36.0075	21.7370	0.6037
2.300	35.9379	21.7004	0.6038
2.305	35.8361	21.7891	0.6080
2.310	35.8498	21.6931	0.6051
2.315	35.8421	21.7781	0.6076
2.320	35.7707	21.7936	0.6093
2.325	35.6867	21.7355	0.6091
2.330	35.6938	21.8713	0.6127
2.335	35.6214	21.8209	0.6126
2.340	35.6491	21.7803	0.6110
2.345	35.5197	21.8335	0.6147
2.350	35.4930	21.8605	0.6159
2.355	35.4618	21.7808	0.6142
2.360	35.4375	21.8676	0.6171
2.365	35.3987	21.8833	0.6182
2.370	35.3741	21.8575	0.6179
2.375	35.3408	21.8137	0.6172
2.380	35.3015	21.8945	0.6202
2.385	35.2333	21.8921	0.6213

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	35.2356	21.9061	0.6217
2.395	35.1402	21.9500	0.6246
2.400	35.0827	21.9329	0.6252
2.405	35.0663	21.8775	0.6239
2.410	35.0601	21.9221	0.6253
2.415	34.9857	21.9198	0.6265
2.420	34.9576	21.9717	0.6285
2.425	34.9083	21.9119	0.6277
2.430	34.8785	21.9068	0.6281
2.435	34.8902	21.9247	0.6284
2.440	34.8242	21.9384	0.6300
2.445	34.7985	21.9588	0.6310
2.450	34.7861	21.9640	0.6314
2.455	34.7235	21.9445	0.6320
2.460	34.6966	21.9666	0.6331
2.465	34.6843	22.0177	0.6348
2.470	34.6061	21.9671	0.6348
2.475	34.5883	21.9806	0.6355
2.480	34.5430	21.9894	0.6366
2.485	34.5315	22.0306	0.6380
2.490	34.4836	22.0080	0.6382
2.495	34.4256	22.0329	0.6400
2.500	34.4409	22.0268	0.6396
2.505	34.3876	22.0099	0.6401
2.510	34.3550	22.0354	0.6414
2.515	34.3062	22.0317	0.6422
2.520	34.2745	22.0588	0.6436
2.525	34.1825	22.0383	0.6447
2.530	34.1778	22.0538	0.6453
2.535	34.1940	22.0625	0.6452
2.540	34.1227	22.0330	0.6457
2.545	34.0932	22.0231	0.6460
2.550	34.0362	22.0516	0.6479
2.555	34.0275	22.0491	0.6480
2.560	33.9638	22.0728	0.6499
2.565	33.9197	22.0596	0.6503
2.570	33.8965	22.0545	0.6506
2.575	33.8651	22.0613	0.6514
2.580	33.7912	22.0418	0.6523
2.585	33.7422	22.1095	0.6552
2.590	33.7218	22.0356	0.6535
2.595	33.6770	22.0761	0.6555
2.600	33.6097	22.0544	0.6562
2.605	33.5341	22.0296	0.6569
2.610	33.5837	22.0579	0.6568
2.615	33.5249	22.0100	0.6565

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	33.4483	22.0508	0.6593
2.625	33.4029	22.0396	0.6598
2.630	33.3920	22.0715	0.6610
2.635	33.3815	22.0370	0.6602
2.640	33.2802	22.0508	0.6626
2.645	33.2887	22.0400	0.6621
2.650	33.2049	22.0643	0.6645
2.655	33.2217	22.0325	0.6632
2.660	33.1512	22.0544	0.6653
2.665	33.1898	22.0491	0.6643
2.670	33.1374	22.0203	0.6645
2.675	33.0814	22.0398	0.6662
2.680	33.0529	22.0551	0.6673
2.685	32.9864	22.0538	0.6686
2.690	32.9532	22.0642	0.6696
2.695	32.9233	22.0902	0.6710
2.700	32.8886	22.1099	0.6723
2.705	32.8622	22.1148	0.6730
2.710	32.8190	22.1703	0.6755
2.715	32.8049	22.1274	0.6745
2.720	32.7333	22.1634	0.6771
2.725	32.6536	22.1468	0.6782
2.730	32.5875	22.1656	0.6802
2.735	32.5776	22.2147	0.6819
2.740	32.5746	22.1237	0.6792
2.745	32.5246	22.2353	0.6836
2.750	32.4628	22.2358	0.6850
2.755	32.4137	22.2334	0.6859
2.760	32.4323	22.2470	0.6860
2.765	32.3544	22.2094	0.6864
2.770	32.3610	22.2919	0.6889
2.775	32.2757	22.2252	0.6886
2.780	32.2706	22.2321	0.6889
2.785	32.2249	22.2670	0.6910
2.790	32.2187	22.2315	0.6900
2.795	32.1667	22.2858	0.6928
2.800	32.1304	22.2711	0.6931
2.805	32.1111	22.2804	0.6939
2.810	32.0251	22.2580	0.6950
2.815	32.0159	22.2872	0.6961
2.820	31.9971	22.2731	0.6961
2.825	31.9390	22.2537	0.6968
2.830	31.9190	22.2894	0.6983
2.835	31.9098	22.2847	0.6984
2.840	31.8271	22.3160	0.7012
2.845	31.7938	22.3166	0.7019

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	31.7777	22.2395	0.6998
2.855	31.7499	22.3083	0.7026
2.860	31.6882	22.2736	0.7029
2.865	31.6226	22.3134	0.7056
2.870	31.6000	22.3005	0.7057
2.875	31.5794	22.2858	0.7057
2.880	31.5384	22.3348	0.7082
2.885	31.5383	22.3269	0.7079
2.890	31.4430	22.3948	0.7122
2.895	31.4400	22.3184	0.7099
2.900	31.4377	22.3239	0.7101
2.905	31.3808	22.3865	0.7134
2.910	31.3642	22.4078	0.7144
2.915	31.3348	22.3659	0.7138
2.920	31.3193	22.3763	0.7145
2.925	31.3168	22.3160	0.7126
2.930	31.2700	22.3361	0.7143
2.935	31.2146	22.3421	0.7158
2.940	31.2199	22.3474	0.7158
2.945	31.1536	22.3569	0.7176
2.950	31.0989	22.3375	0.7183
2.955	31.1819	22.3239	0.7159
2.960	30.9421	22.3924	0.7237
2.965	31.0449	22.2954	0.7182
2.970	30.9418	22.2721	0.7198
2.975	30.9165	22.2879	0.7209
2.980	30.9378	22.3240	0.7216
2.985	30.8438	22.3092	0.7233
2.990	30.7961	22.3608	0.7261
2.995	30.7892	22.2745	0.7235
3.000	30.6908	22.3330	0.7277
3.005	30.7334	22.3108	0.7259
3.010	30.6897	22.2276	0.7243
3.015	30.6404	22.2578	0.7264
3.020	30.5714	22.3652	0.7316
3.025	30.5388	22.3038	0.7303
3.030	30.5552	22.2819	0.7292
3.035	30.5419	22.3630	0.7322
3.040	30.4534	22.3614	0.7343
3.045	30.4073	22.3451	0.7349
3.050	30.3266	22.3073	0.7356
3.055	30.3785	22.4392	0.7387
3.060	30.3479	22.4513	0.7398
3.065	30.4081	22.3847	0.7361
3.070	30.2977	22.4747	0.7418
3.075	30.3287	22.4704	0.7409

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	30.3104	22.4830	0.7418
3.085	30.2493	22.4115	0.7409
3.090	30.2765	22.4525	0.7416
3.095	30.2687	22.4577	0.7419
3.100	30.2709	22.4381	0.7412
3.105	30.2270	22.3739	0.7402
3.110	30.2179	22.4183	0.7419
3.115	30.1890	22.3961	0.7419
3.120	30.1552	22.4170	0.7434
3.125	30.1171	22.3684	0.7427
3.130	30.0833	22.3630	0.7434
3.135	30.0147	22.3469	0.7445
3.140	30.0223	22.3408	0.7441
3.145	29.9432	22.3013	0.7448
3.150	29.9393	22.3203	0.7455
3.155	29.8707	22.3280	0.7475
3.160	29.8822	22.2996	0.7463
3.165	29.7719	22.3227	0.7498
3.170	29.8204	22.2768	0.7470
3.175	29.7085	22.2949	0.7505
3.180	29.7025	22.3077	0.7510
3.185	29.6213	22.3470	0.7544
3.190	29.6294	22.3212	0.7533
3.195	29.5679	22.3624	0.7563
3.200	29.4902	22.3470	0.7578
3.205	29.5151	22.4178	0.7595
3.210	29.4824	22.3853	0.7593
3.215	29.5003	22.3728	0.7584
3.220	29.5388	22.4209	0.7590
3.225	29.4385	22.4510	0.7626
3.230	29.4618	22.3269	0.7578
3.235	29.4140	22.4059	0.7617
3.240	29.3247	22.3833	0.7633
3.245	29.3520	22.3297	0.7608
3.250	29.2186	22.4539	0.7685
3.255	29.1741	22.3493	0.7661
3.260	29.2356	22.3641	0.7650
3.265	29.1865	22.4162	0.7680
3.270	29.1238	22.2649	0.7645
3.275	29.0792	22.3015	0.7669
3.280	29.0967	22.3196	0.7671
3.285	28.9100	22.3460	0.7730
3.290	28.9949	22.2738	0.7682
3.295	28.9064	22.2751	0.7706
3.300	28.9621	22.2759	0.7691
3.305	28.8809	22.3142	0.7726

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	28.8327	22.2801	0.7727
3.315	28.7903	22.2689	0.7735
3.320	28.7864	22.3178	0.7753
3.325	28.7831	22.2822	0.7741
3.330	28.6981	22.3193	0.7777
3.335	28.5952	22.2452	0.7779
3.340	28.6221	22.2829	0.7785
3.345	28.5523	22.2350	0.7787
3.350	28.4775	22.2669	0.7819
3.355	28.5115	22.2534	0.7805
3.360	28.4077	22.3009	0.7850
3.365	28.3837	22.2974	0.7856
3.370	28.3326	22.2965	0.7870
3.375	28.3755	22.3290	0.7869
3.380	28.3445	22.3232	0.7876
3.385	28.2759	22.3225	0.7895
3.390	28.2402	22.3004	0.7897
3.395	28.2452	22.3159	0.7901
3.400	28.1307	22.3500	0.7945
3.405	28.1210	22.3058	0.7932
3.410	28.1279	22.3566	0.7948
3.415	28.0385	22.3327	0.7965
3.420	28.0515	22.3166	0.7956
3.425	28.0685	22.3176	0.7951
3.430	28.0635	22.3222	0.7954
3.435	28.0193	22.3791	0.7987
3.440	28.0062	22.2873	0.7958
3.445	27.9299	22.3184	0.7991
3.450	27.8372	22.2959	0.8009
3.455	27.8939	22.3379	0.8008
3.460	27.8342	22.3072	0.8014
3.465	27.8098	22.2622	0.8005
3.470	27.8031	22.2524	0.8004
3.475	27.7990	22.2064	0.7988
3.480	27.7381	22.1414	0.7982
3.485	27.7616	22.0524	0.7943
3.490	27.6217	22.2239	0.8046
3.495	27.5975	22.2052	0.8046
3.500	27.5646	22.1278	0.8028
3.505	27.6218	22.1942	0.8035
3.510	27.5296	22.0603	0.8013
3.515	27.4439	22.1620	0.8075
3.520	27.3688	22.1085	0.8078
3.525	27.1689	22.2191	0.8178
3.530	27.2334	22.2622	0.8175
3.535	27.1793	22.1897	0.8164

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	27.0826	22.2858	0.8229
3.545	27.1691	22.3960	0.8243
3.550	27.0827	22.4278	0.8281
3.555	27.1424	22.3920	0.8250
3.560	27.1784	22.4848	0.8273
3.565	27.2447	22.4488	0.8240
3.570	27.1298	22.5156	0.8299
3.575	27.2100	22.6780	0.8334
3.580	27.3400	22.6167	0.8272
3.585	27.5207	22.4285	0.8150
3.590	27.5051	22.2233	0.8080
3.595	27.4947	22.1196	0.8045
3.600	27.3931	22.1755	0.8095
3.605	27.3957	21.9792	0.8023
3.610	27.3037	21.9348	0.8034
3.615	27.1386	21.9379	0.8084
3.620	27.0048	21.8971	0.8109
3.625	27.0800	21.7643	0.8037
3.630	26.7367	21.8465	0.8171
3.635	26.7472	21.9555	0.8209
3.640	26.7824	22.0193	0.8222
3.645	26.7068	22.0681	0.8263
3.650	26.6048	22.0958	0.8305
3.655	26.5839	22.0829	0.8307
3.660	26.5594	22.1868	0.8354
3.665	26.5447	22.1328	0.8338
3.670	26.4904	22.2077	0.8383
3.675	26.4898	22.1538	0.8363
3.680	26.5066	22.2286	0.8386
3.685	26.5049	22.2961	0.8412
3.690	26.4858	22.2913	0.8416
3.695	26.5071	22.3573	0.8434
3.700	26.5477	22.3557	0.8421
3.705	26.6007	22.3290	0.8394
3.710	26.5643	22.3372	0.8409
3.715	26.6599	22.3260	0.8374
3.720	26.7291	22.2460	0.8323
3.725	26.6996	22.2455	0.8332
3.730	26.7283	22.1294	0.8279
3.735	26.6669	21.9848	0.8244
3.740	26.6579	21.9150	0.8221
3.745	26.5723	21.8589	0.8226
3.750	26.3966	21.7851	0.8253
3.755	26.2539	21.7928	0.8301
3.760	26.0871	21.8548	0.8378
3.765	25.9259	21.9745	0.8476

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	25.9730	22.1705	0.8536
3.775	25.9971	22.2267	0.8550
3.780	25.7561	22.5279	0.8747
3.785	26.1043	22.3941	0.8579
3.790	26.2358	22.2342	0.8475
3.795	26.2366	22.2018	0.8462
3.800	26.2168	22.2849	0.8500
3.805	26.2611	22.1452	0.8433
3.810	26.1628	22.1072	0.8450
3.815	26.0925	21.9848	0.8426
3.820	26.1320	22.0590	0.8441
3.825	26.0428	21.9627	0.8433
3.830	26.0352	22.0088	0.8453
3.835	25.3931	21.2924	0.8385
3.840	25.9056	21.9838	0.8486
3.845	25.8674	22.0062	0.8507
3.850	25.8519	21.9779	0.8501
3.855	25.7989	22.0114	0.8532
3.860	25.7608	21.9544	0.8522
3.865	25.6676	21.9789	0.8563
3.870	25.7515	21.9766	0.8534
3.875	25.6717	21.9721	0.8559
3.880	25.7001	21.9641	0.8546
3.885	25.6107	21.9781	0.8582
3.890	25.6347	22.0157	0.8588
3.895	25.6247	21.9199	0.8554
3.900	25.5581	21.9309	0.8581
3.905	25.4916	21.9852	0.8625
3.910	25.5195	21.9391	0.8597
3.915	25.4842	21.9355	0.8607
3.920	25.4383	21.9848	0.8642
3.925	25.4063	21.9164	0.8626
3.930	25.3951	21.9146	0.8629
3.935	25.3934	21.9147	0.8630
3.940	25.3388	21.9245	0.8653
3.945	25.3218	21.8846	0.8643
3.950	25.2676	21.8893	0.8663
3.955	25.2298	21.9163	0.8687
3.960	25.2209	21.8884	0.8679
3.965	25.2098	21.8815	0.8680
3.970	25.1637	21.9052	0.8705
3.975	25.1336	21.8749	0.8703
3.980	25.1187	21.8672	0.8706
3.985	25.1078	21.8818	0.8715
3.990	25.0931	21.8743	0.8717
3.995	25.0406	21.8512	0.8726

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	25.0006	21.8622	0.8745
4.005	24.9732	21.8487	0.8749
4.010	24.9288	21.8497	0.8765
4.015	24.9295	21.8746	0.8775
4.020	24.8966	21.8362	0.8771
4.025	24.9361	21.8176	0.8749
4.030	24.8295	21.8199	0.8788
4.035	24.7953	21.8454	0.8810
4.040	24.8136	21.8264	0.8796
4.045	24.7953	21.7881	0.8787
4.050	24.7453	21.7742	0.8799
4.055	24.7083	21.7616	0.8807
4.060	24.6598	21.7851	0.8834
4.065	24.6188	21.7998	0.8855
4.070	24.6293	21.8102	0.8855
4.075	24.5919	21.7786	0.8856
4.080	24.5634	21.8239	0.8885
4.085	24.5458	21.7930	0.8879
4.090	24.5431	21.7796	0.8874
4.095	24.4742	21.7981	0.8907
4.100	24.4891	21.8270	0.8913
4.105	24.4190	21.6737	0.8876
4.110	24.4405	21.7765	0.8910
4.115	24.3880	21.7643	0.8924
4.120	24.3336	21.7709	0.8947
4.125	24.3944	21.7552	0.8918
4.130	24.3654	21.7971	0.8946
4.135	24.2520	21.7224	0.8957
4.140	24.2750	21.6543	0.8920
4.145	24.3540	21.6408	0.8886
4.150	24.1681	21.6885	0.8974
4.155	24.1386	21.6958	0.8988
4.160	24.1959	21.6798	0.8960
4.165	24.1191	21.6170	0.8963
4.170	24.0939	21.6029	0.8966
4.175	24.1287	21.5489	0.8931
4.180	24.1312	21.5987	0.8951
4.185	24.1030	21.5296	0.8932
4.190	24.0481	21.5367	0.8956
4.195	24.0389	21.5427	0.8962
4.200	24.0732	21.5557	0.8954
4.205	24.0605	21.5634	0.8962
4.210	23.9451	21.5711	0.9009
4.215	24.0073	21.5847	0.8991
4.220	23.9782	21.5903	0.9004
4.225	24.0528	21.5920	0.8977

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	23.9720	21.5884	0.9006
4.235	23.9182	21.5865	0.9025
4.240	23.9119	21.6749	0.9064
4.245	23.9511	21.6157	0.9025
4.250	23.8999	21.5884	0.9033
4.255	23.9172	21.6387	0.9047
4.260	23.8381	21.6243	0.9071
4.265	23.9015	21.6150	0.9043
4.270	23.8204	21.6194	0.9076
4.275	23.7926	21.6218	0.9088
4.280	23.7840	21.6443	0.9100
4.285	23.7541	21.6586	0.9118
4.290	23.7300	21.6446	0.9121
4.295	23.7110	21.6376	0.9126
4.300	23.7054	21.6642	0.9139
4.305	23.6718	21.6535	0.9147
4.310	23.6749	21.6440	0.9142
4.315	23.6578	21.6447	0.9149
4.320	23.6545	21.6327	0.9145
4.325	23.6315	21.6361	0.9156
4.330	23.6304	21.6326	0.9155
4.335	23.6120	21.6369	0.9163
4.340	23.6075	21.6020	0.9150
4.345	23.5761	21.5977	0.9161
4.350	23.5605	21.5898	0.9164
4.355	23.5296	21.6047	0.9182
4.360	23.4946	21.5663	0.9179
4.365	23.4421	21.5658	0.9200
4.370	23.4789	21.5594	0.9182
4.375	23.3839	21.5802	0.9229
4.380	23.3921	21.5631	0.9218
4.385	23.2924	21.5483	0.9251
4.390	23.3255	21.5066	0.9220
4.395	23.2623	21.5279	0.9254
4.400	23.2331	21.5469	0.9274
4.405	23.1739	21.5413	0.9296
4.410	23.1916	21.5521	0.9293
4.415	23.1628	21.5098	0.9286
4.420	23.1308	21.5757	0.9328
4.425	23.1593	21.5850	0.9320
4.430	23.0976	21.5205	0.9317
4.435	23.0350	21.4862	0.9328
4.440	23.0562	21.5137	0.9331
4.445	22.9614	21.4778	0.9354
4.450	22.9844	21.4341	0.9326
4.455	22.8992	21.4916	0.9385

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	22.9258	21.3751	0.9324
4.465	22.9235	21.3089	0.9296
4.470	22.9493	21.3379	0.9298
4.475	22.9650	21.2880	0.9270
4.480	22.9944	21.3090	0.9267
4.485	22.8685	21.2507	0.9293
4.490	22.8919	21.2900	0.9300
4.495	22.8860	21.2784	0.9298
4.500	22.8406	21.2836	0.9318
4.505	22.7877	21.2682	0.9333
4.510	22.6907	21.2726	0.9375
4.515	22.7070	21.3566	0.9405
4.520	22.6865	21.3490	0.9410
4.525	22.7211	21.3311	0.9388
4.530	22.6282	21.3035	0.9415
4.535	22.6314	21.3354	0.9427
4.540	22.6204	21.2958	0.9414
4.545	22.5655	21.2840	0.9432
4.550	22.5568	21.2403	0.9416
4.555	22.5206	21.2769	0.9448
4.560	22.5264	21.2748	0.9444
4.565	22.4964	21.2949	0.9466
4.570	22.4820	21.2557	0.9455
4.575	22.4669	21.2616	0.9464
4.580	22.4631	21.2855	0.9476
4.585	22.4275	21.2513	0.9476
4.590	22.4059	21.2304	0.9475
4.595	22.3762	21.2546	0.9499
4.600	22.3576	21.2369	0.9499
4.605	22.3204	21.2261	0.9510
4.610	22.3103	21.2347	0.9518
4.615	22.2848	21.2082	0.9517
4.620	22.2695	21.1670	0.9505
4.625	22.2616	21.1930	0.9520
4.630	22.2443	21.1706	0.9517
4.635	22.2060	21.1437	0.9522
4.640	22.1688	21.1469	0.9539
4.645	22.1747	21.1195	0.9524
4.650	22.1470	21.1274	0.9540
4.655	22.1136	21.0804	0.9533
4.660	22.1457	21.1111	0.9533
4.665	22.1137	21.0930	0.9538
4.670	22.0076	21.0863	0.9581
4.675	22.0061	21.0048	0.9545
4.680	22.0059	21.0331	0.9558
4.685	21.9971	21.0196	0.9556

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	21.9819	21.0831	0.9591
4.695	21.9883	20.9869	0.9545
4.700	21.9332	21.0571	0.9601
4.705	21.8625	20.8969	0.9558
4.710	21.7674	21.0477	0.9669
4.715	21.8286	21.0461	0.9642
4.720	21.8089	20.9823	0.9621
4.725	21.8146	21.0638	0.9656
4.730	21.8305	21.0175	0.9628
4.735	21.8329	21.0640	0.9648
4.740	21.8313	21.0376	0.9636
4.745	21.7351	21.0434	0.9682
4.750	21.8183	21.0024	0.9626
4.755	21.8000	21.0580	0.9660
4.760	21.7488	20.9922	0.9652
4.765	21.8273	21.0142	0.9628
4.770	21.7892	21.0420	0.9657
4.775	21.6504	20.9210	0.9663
4.780	21.6872	20.9286	0.9650
4.785	21.6911	21.1329	0.9743
4.790	21.6588	20.8939	0.9647
4.795	21.5708	20.9118	0.9695
4.800	21.5873	20.9213	0.9691
4.805	21.6126	20.8435	0.9644
4.810	21.5540	20.9645	0.9726
4.815	21.5782	20.8429	0.9659
4.820	21.4574	20.9421	0.9760
4.825	21.5067	20.9346	0.9734
4.830	21.4321	20.9451	0.9773
4.835	21.4385	20.9629	0.9778
4.840	21.4565	20.9482	0.9763
4.845	21.4795	20.9130	0.9736
4.850	21.3812	20.9428	0.9795
4.855	21.4618	20.9633	0.9768
4.860	21.4246	20.9255	0.9767
4.865	21.4640	20.8978	0.9736
4.870	21.4026	20.9075	0.9769
4.875	21.4125	20.9334	0.9776
4.880	21.4084	20.9573	0.9789
4.885	21.3758	20.8790	0.9768
4.890	21.4527	20.9420	0.9762
4.895	21.3529	20.9003	0.9788
4.900	21.3751	20.9314	0.9792
4.905	21.3601	20.9117	0.9790
4.910	21.3185	20.9017	0.9805
4.915	21.3124	20.9072	0.9810

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	21.2990	20.8972	0.9811
4.925	21.2918	20.8878	0.9810
4.930	21.2461	20.8773	0.9826
4.935	21.2394	20.8751	0.9828
4.940	21.2345	20.8740	0.9830
4.945	21.1973	20.8849	0.9853
4.950	21.1940	20.8762	0.9850
4.955	21.1829	20.8938	0.9864
4.960	21.1484	20.8663	0.9867
4.965	21.1381	20.8686	0.9872
4.970	21.1127	20.8843	0.9892
4.975	21.1249	20.8701	0.9879
4.980	21.0445	20.8797	0.9922
4.985	21.0207	20.8552	0.9921
4.990	20.9939	20.8762	0.9944
4.995	20.9099	20.8769	0.9984
5.000	20.8918	20.8454	0.9978
5.005	20.8455	20.8431	0.9999
5.010	20.7997	20.8092	1.0005
5.015	20.7716	20.8277	1.0027
5.020	20.7321	20.7657	1.0016
5.025	20.6446	20.7427	1.0048
5.030	20.5924	20.6972	1.0051
5.035	20.6300	20.7088	1.0038
5.040	20.5810	20.6549	1.0036
5.045	20.6027	20.5994	0.9998
5.050	20.5708	20.5173	0.9974
5.055	20.6331	20.5428	0.9956
5.060	20.5965	20.5201	0.9963
5.065	20.6209	20.4775	0.9930
5.070	20.6130	20.5432	0.9966
5.075	20.6074	20.5238	0.9959
5.080	20.5748	20.5040	0.9966
5.085	20.5638	20.4964	0.9967
5.090	20.5792	20.5384	0.9980
5.095	20.6013	20.5689	0.9984
5.100	20.5799	20.5644	0.9992
5.105	20.5494	20.6570	1.0052
5.110	20.5091	20.5530	1.0021
5.115	20.4704	20.5478	1.0038
5.120	20.4510	20.5494	1.0048
5.125	20.4354	20.5681	1.0065
5.130	20.4102	20.5295	1.0058
5.135	20.3541	20.5222	1.0083
5.140	20.3476	20.5436	1.0096
5.145	20.3007	20.4631	1.0080

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	20.2954	20.5237	1.0112
5.155	20.2668	20.4775	1.0104
5.160	20.2529	20.4633	1.0104
5.165	20.2143	20.4609	1.0122
5.170	20.1987	20.4259	1.0112
5.175	20.1660	20.4022	1.0117
5.180	20.1697	20.4048	1.0117
5.185	20.1381	20.4002	1.0130
5.190	20.1433	20.3754	1.0115
5.195	20.1362	20.3669	1.0115
5.200	20.1414	20.3717	1.0114
5.205	20.1360	20.3800	1.0121
5.210	20.1207	20.3558	1.0117
5.215	20.1119	20.3619	1.0124
5.220	20.0811	20.3422	1.0130
5.225	20.0456	20.3405	1.0147
5.230	20.0689	20.3269	1.0129
5.235	20.0264	20.3501	1.0162
5.240	20.0064	20.3254	1.0159
5.245	20.0166	20.3329	1.0158
5.250	19.9770	20.3497	1.0187
5.255	19.9229	20.3468	1.0213
5.260	19.9981	20.3092	1.0156
5.265	19.9876	20.3677	1.0190
5.270	19.9280	20.3144	1.0194
5.275	19.8958	20.2773	1.0192
5.280	19.8941	20.3083	1.0208
5.285	19.9152	20.2808	1.0184
5.290	19.8655	20.2510	1.0194
5.295	19.8992	20.1904	1.0146
5.300	19.8459	20.2427	1.0200
5.305	19.8815	20.2308	1.0176
5.310	19.7488	20.2199	1.0239
5.315	19.8022	20.2262	1.0214
5.320	19.7525	20.1682	1.0210
5.325	19.7717	20.2053	1.0219
5.330	19.7409	20.2023	1.0234
5.335	19.6552	20.2106	1.0283
5.340	19.6484	20.2515	1.0307
5.345	19.6237	20.2769	1.0333
5.350	19.7032	20.1846	1.0244
5.355	19.6080	20.2095	1.0307
5.360	19.5962	20.1132	1.0264
5.365	19.6926	20.1894	1.0252
5.370	19.5633	20.1193	1.0284
5.375	19.5939	20.1800	1.0299

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	19.6001	20.1777	1.0295
5.385	19.5256	20.1280	1.0309
5.390	19.5206	20.1973	1.0347
5.395	19.5707	20.1849	1.0314
5.400	19.5014	20.1631	1.0339
5.405	19.5025	20.1442	1.0329
5.410	19.4714	20.1620	1.0355
5.415	19.4821	20.1106	1.0323
5.420	19.4266	20.1617	1.0378
5.425	19.4162	20.1680	1.0387
5.430	19.4401	20.1165	1.0348
5.435	19.4227	20.1095	1.0354
5.440	19.3465	20.1514	1.0416
5.445	19.4081	20.1344	1.0374
5.450	19.3290	20.1159	1.0407
5.455	19.3737	20.1249	1.0388
5.460	19.3348	20.1151	1.0404
5.465	19.3559	20.1667	1.0419
5.470	19.2568	20.1332	1.0455
5.475	19.2937	20.1103	1.0423
5.480	19.2599	20.1436	1.0459
5.485	19.2690	20.1044	1.0434
5.490	19.2062	20.1587	1.0496
5.495	19.2640	20.1097	1.0439
5.500	19.2367	20.1406	1.0470
5.505	19.2374	20.1444	1.0471
5.510	19.2346	20.1237	1.0462
5.515	19.2390	20.1016	1.0448
5.520	19.1958	20.0942	1.0468
5.525	19.2151	20.0921	1.0456
5.530	19.1944	20.0860	1.0465
5.535	19.2052	20.0669	1.0449
5.540	19.1685	20.0887	1.0480
5.545	19.1361	20.0545	1.0480
5.550	19.1460	20.0608	1.0478
5.555	19.1458	20.0408	1.0467
5.560	19.1314	20.0418	1.0476
5.565	19.1440	20.0403	1.0468
5.570	19.1344	20.0429	1.0475
5.575	19.1408	20.0394	1.0469
5.580	19.1335	20.0630	1.0486
5.585	19.1154	20.0457	1.0487
5.590	19.1347	20.0187	1.0462
5.595	19.1127	20.0116	1.0470
5.600	19.1228	20.0124	1.0465
5.605	19.1388	19.9755	1.0437

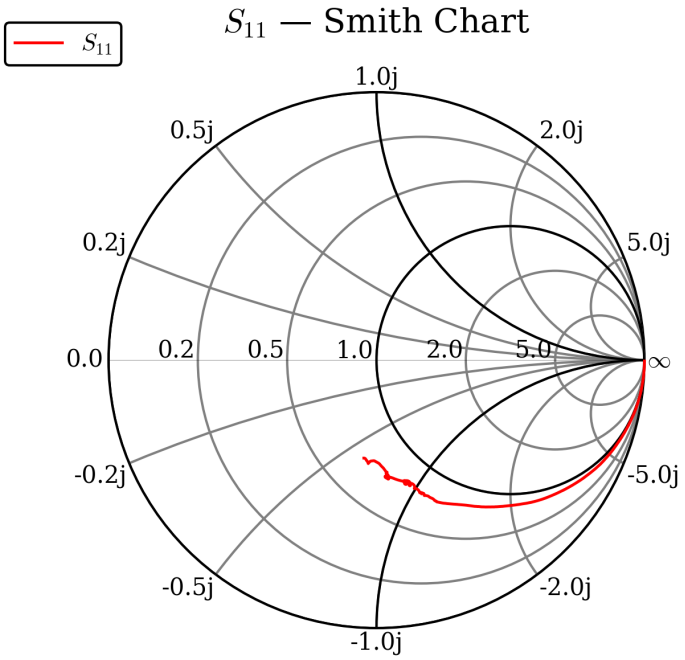
(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	19.0985	19.9427	1.0442
5.615	19.0852	19.9047	1.0429
5.620	19.0670	19.9098	1.0442
5.625	19.0588	19.9267	1.0455
5.630	19.0246	19.8769	1.0448
5.635	19.0274	19.8566	1.0436
5.640	19.0080	19.8100	1.0422
5.645	18.9632	19.8028	1.0443
5.650	18.9726	19.8434	1.0459
5.655	18.9339	19.8159	1.0466
5.660	18.9074	19.8236	1.0485
5.665	18.9807	19.8034	1.0433
5.670	18.8201	19.8229	1.0533
5.675	18.8687	19.7545	1.0469
5.680	18.8217	19.7642	1.0501
5.685	18.8820	19.7295	1.0449
5.690	18.8340	19.7291	1.0475
5.695	18.8152	19.7755	1.0510
5.700	18.7756	19.6846	1.0484
5.705	18.6988	19.6971	1.0534
5.710	18.7747	19.6714	1.0478
5.715	18.7091	19.6236	1.0489
5.720	18.6998	19.6516	1.0509
5.725	18.6435	19.6607	1.0546
5.730	18.6527	19.6280	1.0523
5.735	18.6328	19.6288	1.0535
5.740	18.6082	19.6372	1.0553
5.745	18.6016	19.6628	1.0570
5.750	18.5516	19.6086	1.0570
5.755	18.5716	19.6104	1.0559
5.760	18.5030	19.5763	1.0580
5.765	18.5624	19.5594	1.0537
5.770	18.5280	19.5344	1.0543
5.775	18.5166	19.5662	1.0567
5.780	18.4537	19.5737	1.0607
5.785	18.4346	19.5443	1.0602
5.790	18.4045	19.5728	1.0635
5.795	18.4188	19.5782	1.0629
5.800	18.3836	19.5544	1.0637
5.805	18.3599	19.5353	1.0640
5.810	18.3561	19.5584	1.0655
5.815	18.3500	19.5528	1.0655
5.820	18.3350	19.5340	1.0654
5.825	18.3231	19.5181	1.0652
5.830	18.2761	19.5239	1.0683
5.835	18.2794	19.5182	1.0678

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	18.2629	19.5134	1.0685
5.845	18.2474	19.5303	1.0703
5.850	18.2434	19.4957	1.0686
5.855	18.2161	19.4904	1.0700
5.860	18.1929	19.4937	1.0715
5.865	18.1910	19.4983	1.0719
5.870	18.1775	19.4625	1.0707
5.875	18.1594	19.4675	1.0720
5.880	18.1446	19.4516	1.0720
5.885	18.1520	19.4231	1.0700
5.890	18.0749	19.4766	1.0775
5.895	18.0935	19.4348	1.0741
5.900	18.0834	19.3919	1.0724
5.905	18.0603	19.4538	1.0772
5.910	18.0380	19.3660	1.0736
5.915	18.0222	19.3995	1.0764
5.920	18.0150	19.3748	1.0755
5.925	17.9917	19.3543	1.0757
5.930	17.9255	19.3866	1.0815
5.935	17.9215	19.3614	1.0803
5.940	17.9152	19.3618	1.0807
5.945	17.9764	19.3356	1.0756
5.950	17.9029	19.3731	1.0821
5.955	17.9045	19.3810	1.0825
5.960	17.8802	19.3557	1.0825
5.965	17.8700	19.3447	1.0825
5.970	17.8567	19.3813	1.0854
5.975	17.8999	19.2519	1.0755
5.980	17.9718	19.3249	1.0753
5.985	17.8029	19.3465	1.0867
5.990	17.7920	19.2799	1.0836
5.995	17.7990	19.2673	1.0825
6.000	17.8450	19.2483	1.0786

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0036	0.00
0.010	0.9999	-0.0071	-0.00
0.015	0.9999	-0.0107	-0.00
0.020	0.9997	-0.0142	-0.00
0.025	0.9997	-0.0176	-0.00
0.030	0.9996	-0.0213	-0.00
0.035	0.9993	-0.0247	-0.00
0.040	0.9991	-0.0284	-0.00
0.045	0.9990	-0.0318	-0.00
0.050	0.9988	-0.0353	-0.00
0.055	0.9985	-0.0387	-0.01
0.060	0.9982	-0.0424	-0.01
0.065	0.9980	-0.0459	-0.01
0.070	0.9977	-0.0493	-0.01
0.075	0.9973	-0.0528	-0.01
0.080	0.9970	-0.0562	-0.01
0.085	0.9965	-0.0598	-0.01
0.090	0.9962	-0.0634	-0.02
0.095	0.9957	-0.0668	-0.02
0.100	0.9953	-0.0703	-0.02
0.105	0.9947	-0.0736	-0.02
0.110	0.9942	-0.0773	-0.02
0.115	0.9938	-0.0807	-0.03

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9931	-0.0841	-0.03
0.125	0.9926	-0.0876	-0.03
0.130	0.9919	-0.0911	-0.03
0.135	0.9914	-0.0944	-0.04
0.140	0.9909	-0.0979	-0.04
0.145	0.9900	-0.1014	-0.04
0.150	0.9895	-0.1047	-0.04
0.155	0.9888	-0.1080	-0.05
0.160	0.9879	-0.1115	-0.05
0.165	0.9874	-0.1148	-0.05
0.170	0.9865	-0.1182	-0.06
0.175	0.9858	-0.1216	-0.06
0.180	0.9849	-0.1250	-0.06
0.185	0.9841	-0.1282	-0.07
0.190	0.9832	-0.1317	-0.07
0.195	0.9825	-0.1351	-0.07
0.200	0.9815	-0.1384	-0.08
0.205	0.9806	-0.1418	-0.08
0.210	0.9796	-0.1451	-0.08
0.215	0.9785	-0.1483	-0.09
0.220	0.9776	-0.1517	-0.09
0.225	0.9766	-0.1551	-0.10
0.230	0.9757	-0.1584	-0.10
0.235	0.9745	-0.1617	-0.11
0.240	0.9735	-0.1650	-0.11
0.245	0.9723	-0.1683	-0.12
0.250	0.9713	-0.1715	-0.12
0.255	0.9701	-0.1747	-0.12
0.260	0.9690	-0.1780	-0.13
0.265	0.9678	-0.1812	-0.13
0.270	0.9666	-0.1845	-0.14
0.275	0.9654	-0.1876	-0.14
0.280	0.9641	-0.1909	-0.15
0.285	0.9630	-0.1941	-0.15
0.290	0.9617	-0.1972	-0.16
0.295	0.9604	-0.2003	-0.17
0.300	0.9591	-0.2034	-0.17
0.305	0.9577	-0.2066	-0.18
0.310	0.9564	-0.2097	-0.18
0.315	0.9549	-0.2128	-0.19
0.320	0.9536	-0.2159	-0.20
0.325	0.9521	-0.2189	-0.20
0.330	0.9508	-0.2220	-0.21
0.335	0.9493	-0.2250	-0.21
0.340	0.9480	-0.2280	-0.22
0.345	0.9465	-0.2312	-0.23

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9449	-0.2341	-0.23
0.355	0.9436	-0.2371	-0.24
0.360	0.9421	-0.2400	-0.24
0.365	0.9405	-0.2430	-0.25
0.370	0.9389	-0.2458	-0.26
0.375	0.9373	-0.2489	-0.27
0.380	0.9358	-0.2517	-0.27
0.385	0.9341	-0.2546	-0.28
0.390	0.9326	-0.2575	-0.29
0.395	0.9311	-0.2603	-0.29
0.400	0.9295	-0.2632	-0.30
0.405	0.9279	-0.2661	-0.31
0.410	0.9264	-0.2688	-0.31
0.415	0.9248	-0.2714	-0.32
0.420	0.9231	-0.2743	-0.33
0.425	0.9214	-0.2770	-0.34
0.430	0.9198	-0.2797	-0.34
0.435	0.9182	-0.2824	-0.35
0.440	0.9166	-0.2850	-0.36
0.445	0.9148	-0.2877	-0.36
0.450	0.9132	-0.2904	-0.37
0.455	0.9115	-0.2929	-0.38
0.460	0.9099	-0.2956	-0.38
0.465	0.9081	-0.2982	-0.39
0.470	0.9062	-0.3008	-0.40
0.475	0.9044	-0.3033	-0.41
0.480	0.9030	-0.3059	-0.41
0.485	0.9011	-0.3083	-0.42
0.490	0.8994	-0.3108	-0.43
0.495	0.8977	-0.3133	-0.44
0.500	0.8959	-0.3158	-0.45
0.505	0.8942	-0.3182	-0.45
0.510	0.8925	-0.3207	-0.46
0.515	0.8908	-0.3232	-0.47
0.520	0.8891	-0.3255	-0.47
0.525	0.8873	-0.3279	-0.48
0.530	0.8856	-0.3302	-0.49
0.535	0.8840	-0.3325	-0.50
0.540	0.8821	-0.3349	-0.50
0.545	0.8803	-0.3373	-0.51
0.550	0.8785	-0.3396	-0.52
0.555	0.8767	-0.3418	-0.53
0.560	0.8748	-0.3441	-0.54
0.565	0.8731	-0.3463	-0.54
0.570	0.8713	-0.3485	-0.55
0.575	0.8694	-0.3509	-0.56

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8676	-0.3530	-0.57
0.585	0.8658	-0.3552	-0.58
0.590	0.8639	-0.3573	-0.58
0.595	0.8622	-0.3596	-0.59
0.600	0.8604	-0.3616	-0.60
0.605	0.8585	-0.3638	-0.61
0.610	0.8565	-0.3657	-0.62
0.615	0.8547	-0.3679	-0.63
0.620	0.8528	-0.3700	-0.63
0.625	0.8510	-0.3721	-0.64
0.630	0.8491	-0.3741	-0.65
0.635	0.8472	-0.3761	-0.66
0.640	0.8453	-0.3781	-0.67
0.645	0.8435	-0.3801	-0.68
0.650	0.8416	-0.3820	-0.68
0.655	0.8397	-0.3840	-0.69
0.660	0.8379	-0.3859	-0.70
0.665	0.8359	-0.3878	-0.71
0.670	0.8339	-0.3897	-0.72
0.675	0.8321	-0.3916	-0.73
0.680	0.8302	-0.3935	-0.74
0.685	0.8282	-0.3952	-0.75
0.690	0.8263	-0.3970	-0.76
0.695	0.8242	-0.3987	-0.77
0.700	0.8222	-0.4006	-0.78
0.705	0.8203	-0.4023	-0.78
0.710	0.8182	-0.4041	-0.79
0.715	0.8163	-0.4058	-0.80
0.720	0.8143	-0.4074	-0.81
0.725	0.8123	-0.4092	-0.82
0.730	0.8103	-0.4109	-0.83
0.735	0.8082	-0.4125	-0.84
0.740	0.8062	-0.4141	-0.85
0.745	0.8044	-0.4157	-0.86
0.750	0.8025	-0.4173	-0.87
0.755	0.8004	-0.4189	-0.88
0.760	0.7984	-0.4204	-0.89
0.765	0.7964	-0.4221	-0.90
0.770	0.7943	-0.4236	-0.91
0.775	0.7922	-0.4252	-0.92
0.780	0.7902	-0.4268	-0.93
0.785	0.7883	-0.4285	-0.94
0.790	0.7861	-0.4300	-0.95
0.795	0.7841	-0.4316	-0.96
0.800	0.7820	-0.4330	-0.97
0.805	0.7800	-0.4345	-0.98

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.810	0.7779	-0.4362	-0.99
0.815	0.7759	-0.4376	-1.00
0.820	0.7737	-0.4391	-1.02
0.825	0.7717	-0.4406	-1.03
0.830	0.7696	-0.4422	-1.04
0.835	0.7674	-0.4436	-1.05
0.840	0.7653	-0.4451	-1.06
0.845	0.7632	-0.4465	-1.07
0.850	0.7612	-0.4479	-1.08
0.855	0.7589	-0.4494	-1.09
0.860	0.7568	-0.4508	-1.10
0.865	0.7546	-0.4522	-1.11
0.870	0.7524	-0.4535	-1.12
0.875	0.7503	-0.4548	-1.14
0.880	0.7482	-0.4563	-1.15
0.885	0.7460	-0.4576	-1.16
0.890	0.7438	-0.4589	-1.17
0.895	0.7417	-0.4604	-1.18
0.900	0.7395	-0.4617	-1.19
0.905	0.7373	-0.4630	-1.20
0.910	0.7351	-0.4642	-1.22
0.915	0.7330	-0.4657	-1.23
0.920	0.7307	-0.4669	-1.24
0.925	0.7285	-0.4681	-1.25
0.930	0.7262	-0.4694	-1.26
0.935	0.7241	-0.4707	-1.27
0.940	0.7218	-0.4721	-1.28
0.945	0.7197	-0.4732	-1.30
0.950	0.7175	-0.4746	-1.31
0.955	0.7151	-0.4758	-1.32
0.960	0.7129	-0.4770	-1.33
0.965	0.7107	-0.4782	-1.34
0.970	0.7085	-0.4795	-1.36
0.975	0.7064	-0.4806	-1.37
0.980	0.7042	-0.4818	-1.38
0.985	0.7019	-0.4829	-1.39
0.990	0.6998	-0.4840	-1.40
0.995	0.6977	-0.4852	-1.41
1.000	0.6953	-0.4863	-1.43
1.005	0.6931	-0.4874	-1.44
1.010	0.6909	-0.4884	-1.45
1.015	0.6887	-0.4895	-1.46
1.020	0.6865	-0.4905	-1.48
1.025	0.6843	-0.4917	-1.49
1.030	0.6822	-0.4927	-1.50
1.035	0.6800	-0.4938	-1.51

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.6778	-0.4948	-1.52
1.045	0.6757	-0.4957	-1.53
1.050	0.6736	-0.4967	-1.55
1.055	0.6715	-0.4976	-1.56
1.060	0.6695	-0.4986	-1.57
1.065	0.6673	-0.4996	-1.58
1.070	0.6651	-0.5006	-1.59
1.075	0.6631	-0.5014	-1.60
1.080	0.6610	-0.5024	-1.62
1.085	0.6590	-0.5033	-1.63
1.090	0.6569	-0.5042	-1.64
1.095	0.6548	-0.5050	-1.65
1.100	0.6529	-0.5059	-1.66
1.105	0.6508	-0.5067	-1.67
1.110	0.6486	-0.5076	-1.69
1.115	0.6467	-0.5084	-1.70
1.120	0.6447	-0.5092	-1.71
1.125	0.6427	-0.5100	-1.72
1.130	0.6408	-0.5108	-1.73
1.135	0.6387	-0.5115	-1.74
1.140	0.6368	-0.5124	-1.75
1.145	0.6348	-0.5130	-1.76
1.150	0.6331	-0.5139	-1.77
1.155	0.6311	-0.5146	-1.78
1.160	0.6291	-0.5152	-1.80
1.165	0.6273	-0.5159	-1.81
1.170	0.6254	-0.5166	-1.82
1.175	0.6237	-0.5174	-1.83
1.180	0.6218	-0.5180	-1.84
1.185	0.6200	-0.5185	-1.85
1.190	0.6183	-0.5192	-1.86
1.195	0.6166	-0.5197	-1.87
1.200	0.6148	-0.5203	-1.88
1.205	0.6130	-0.5208	-1.89
1.210	0.6113	-0.5214	-1.90
1.215	0.6096	-0.5219	-1.91
1.220	0.6078	-0.5224	-1.92
1.225	0.6061	-0.5229	-1.93
1.230	0.6044	-0.5235	-1.94
1.235	0.6027	-0.5239	-1.95
1.240	0.6009	-0.5244	-1.96
1.245	0.5993	-0.5250	-1.97
1.250	0.5975	-0.5254	-1.99
1.255	0.5959	-0.5259	-2.00
1.260	0.5942	-0.5261	-2.01
1.265	0.5927	-0.5267	-2.02

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.5910	-0.5272	-2.03
1.275	0.5894	-0.5276	-2.04
1.280	0.5877	-0.5280	-2.05
1.285	0.5861	-0.5285	-2.06
1.290	0.5844	-0.5288	-2.07
1.295	0.5826	-0.5294	-2.08
1.300	0.5809	-0.5297	-2.09
1.305	0.5795	-0.5302	-2.10
1.310	0.5779	-0.5305	-2.11
1.315	0.5763	-0.5308	-2.12
1.320	0.5748	-0.5313	-2.13
1.325	0.5732	-0.5316	-2.14
1.330	0.5717	-0.5317	-2.15
1.335	0.5701	-0.5321	-2.16
1.340	0.5685	-0.5325	-2.17
1.345	0.5670	-0.5327	-2.18
1.350	0.5653	-0.5330	-2.19
1.355	0.5638	-0.5334	-2.20
1.360	0.5622	-0.5338	-2.21
1.365	0.5606	-0.5338	-2.22
1.370	0.5590	-0.5343	-2.23
1.375	0.5573	-0.5345	-2.25
1.380	0.5557	-0.5347	-2.26
1.385	0.5540	-0.5350	-2.27
1.390	0.5524	-0.5353	-2.28
1.395	0.5507	-0.5355	-2.29
1.400	0.5491	-0.5357	-2.30
1.405	0.5473	-0.5359	-2.32
1.410	0.5455	-0.5362	-2.33
1.415	0.5439	-0.5366	-2.34
1.420	0.5422	-0.5368	-2.35
1.425	0.5404	-0.5372	-2.36
1.430	0.5387	-0.5375	-2.37
1.435	0.5369	-0.5378	-2.39
1.440	0.5351	-0.5381	-2.40
1.445	0.5333	-0.5384	-2.41
1.450	0.5316	-0.5386	-2.42
1.455	0.5298	-0.5388	-2.43
1.460	0.5280	-0.5392	-2.44
1.465	0.5262	-0.5394	-2.46
1.470	0.5245	-0.5396	-2.47
1.475	0.5225	-0.5400	-2.48
1.480	0.5206	-0.5402	-2.50
1.485	0.5187	-0.5404	-2.51
1.490	0.5167	-0.5408	-2.52
1.495	0.5150	-0.5410	-2.53

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.5131	-0.5413	-2.55
1.505	0.5112	-0.5414	-2.56
1.510	0.5093	-0.5415	-2.58
1.515	0.5073	-0.5418	-2.59
1.520	0.5053	-0.5421	-2.60
1.525	0.5033	-0.5423	-2.62
1.530	0.5015	-0.5425	-2.63
1.535	0.4994	-0.5428	-2.64
1.540	0.4974	-0.5429	-2.66
1.545	0.4954	-0.5431	-2.67
1.550	0.4934	-0.5433	-2.69
1.555	0.4913	-0.5435	-2.70
1.560	0.4894	-0.5436	-2.72
1.565	0.4872	-0.5439	-2.73
1.570	0.4852	-0.5442	-2.74
1.575	0.4831	-0.5444	-2.76
1.580	0.4812	-0.5444	-2.77
1.585	0.4791	-0.5447	-2.79
1.590	0.4771	-0.5450	-2.80
1.595	0.4750	-0.5452	-2.82
1.600	0.4730	-0.5453	-2.83
1.605	0.4710	-0.5456	-2.84
1.610	0.4691	-0.5459	-2.86
1.615	0.4669	-0.5458	-2.87
1.620	0.4649	-0.5460	-2.89
1.625	0.4630	-0.5461	-2.90
1.630	0.4609	-0.5463	-2.92
1.635	0.4589	-0.5465	-2.93
1.640	0.4567	-0.5466	-2.95
1.645	0.4547	-0.5467	-2.96
1.650	0.4525	-0.5468	-2.98
1.655	0.4505	-0.5470	-2.99
1.660	0.4487	-0.5472	-3.00
1.665	0.4466	-0.5471	-3.02
1.670	0.4446	-0.5472	-3.04
1.675	0.4427	-0.5475	-3.05
1.680	0.4407	-0.5476	-3.06
1.685	0.4388	-0.5478	-3.07
1.690	0.4369	-0.5477	-3.09
1.695	0.4352	-0.5477	-3.10
1.700	0.4331	-0.5476	-3.12
1.705	0.4313	-0.5477	-3.13
1.710	0.4294	-0.5478	-3.15
1.715	0.4274	-0.5478	-3.16
1.720	0.4255	-0.5479	-3.18
1.725	0.4238	-0.5479	-3.19

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.730	0.4218	-0.5480	-3.20
1.735	0.4200	-0.5480	-3.22
1.740	0.4182	-0.5478	-3.23
1.745	0.4166	-0.5480	-3.24
1.750	0.4148	-0.5480	-3.26
1.755	0.4130	-0.5478	-3.27
1.760	0.4115	-0.5480	-3.28
1.765	0.4100	-0.5481	-3.29
1.770	0.4082	-0.5481	-3.31
1.775	0.4069	-0.5481	-3.32
1.780	0.4055	-0.5480	-3.33
1.785	0.4040	-0.5478	-3.34
1.790	0.4024	-0.5477	-3.35
1.795	0.4008	-0.5477	-3.37
1.800	0.3992	-0.5476	-3.38
1.805	0.3978	-0.5476	-3.39
1.810	0.3964	-0.5476	-3.40
1.815	0.3950	-0.5477	-3.41
1.820	0.3937	-0.5475	-3.42
1.825	0.3924	-0.5476	-3.43
1.830	0.3911	-0.5473	-3.44
1.835	0.3898	-0.5472	-3.45
1.840	0.3885	-0.5473	-3.46
1.845	0.3874	-0.5470	-3.47
1.850	0.3860	-0.5468	-3.49
1.855	0.3847	-0.5467	-3.50
1.860	0.3834	-0.5465	-3.51
1.865	0.3822	-0.5464	-3.52
1.870	0.3810	-0.5463	-3.53
1.875	0.3797	-0.5464	-3.54
1.880	0.3786	-0.5464	-3.55
1.885	0.3776	-0.5465	-3.55
1.890	0.3768	-0.5464	-3.56
1.895	0.3758	-0.5462	-3.57
1.900	0.3748	-0.5459	-3.58
1.905	0.3736	-0.5459	-3.59
1.910	0.3726	-0.5459	-3.60
1.915	0.3716	-0.5459	-3.60
1.920	0.3706	-0.5459	-3.61
1.925	0.3696	-0.5458	-3.62
1.930	0.3686	-0.5458	-3.63
1.935	0.3678	-0.5457	-3.63
1.940	0.3669	-0.5456	-3.64
1.945	0.3660	-0.5456	-3.65
1.950	0.3651	-0.5454	-3.66
1.955	0.3643	-0.5454	-3.66

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.3633	-0.5452	-3.67
1.965	0.3625	-0.5451	-3.68
1.970	0.3617	-0.5450	-3.69
1.975	0.3610	-0.5448	-3.69
1.980	0.3601	-0.5448	-3.70
1.985	0.3592	-0.5445	-3.71
1.990	0.3583	-0.5443	-3.72
1.995	0.3574	-0.5443	-3.73
2.000	0.3567	-0.5442	-3.73
2.005	0.3558	-0.5442	-3.74
2.010	0.3549	-0.5440	-3.75
2.015	0.3541	-0.5439	-3.76
2.020	0.3532	-0.5438	-3.76
2.025	0.3523	-0.5437	-3.77
2.030	0.3514	-0.5436	-3.78
2.035	0.3506	-0.5434	-3.79
2.040	0.3496	-0.5432	-3.80
2.045	0.3487	-0.5431	-3.80
2.050	0.3478	-0.5429	-3.81
2.055	0.3468	-0.5428	-3.82
2.060	0.3459	-0.5427	-3.83
2.065	0.3448	-0.5425	-3.84
2.070	0.3439	-0.5425	-3.85
2.075	0.3428	-0.5422	-3.86
2.080	0.3418	-0.5422	-3.86
2.085	0.3407	-0.5420	-3.87
2.090	0.3396	-0.5417	-3.88
2.095	0.3384	-0.5416	-3.90
2.100	0.3373	-0.5415	-3.90
2.105	0.3361	-0.5413	-3.91
2.110	0.3349	-0.5412	-3.93
2.115	0.3337	-0.5412	-3.93
2.120	0.3325	-0.5412	-3.94
2.125	0.3312	-0.5409	-3.95
2.130	0.3299	-0.5408	-3.96
2.135	0.3286	-0.5407	-3.98
2.140	0.3273	-0.5408	-3.98
2.145	0.3261	-0.5406	-3.99
2.150	0.3247	-0.5404	-4.01
2.155	0.3234	-0.5403	-4.02
2.160	0.3221	-0.5400	-4.03
2.165	0.3206	-0.5401	-4.04
2.170	0.3193	-0.5401	-4.05
2.175	0.3179	-0.5398	-4.06
2.180	0.3164	-0.5396	-4.07
2.185	0.3150	-0.5396	-4.08

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.3136	-0.5394	-4.10
2.195	0.3120	-0.5393	-4.11
2.200	0.3106	-0.5392	-4.12
2.205	0.3091	-0.5390	-4.13
2.210	0.3075	-0.5389	-4.15
2.215	0.3059	-0.5388	-4.16
2.220	0.3044	-0.5387	-4.17
2.225	0.3029	-0.5386	-4.18
2.230	0.3014	-0.5382	-4.20
2.235	0.2998	-0.5383	-4.21
2.240	0.2981	-0.5382	-4.22
2.245	0.2966	-0.5380	-4.23
2.250	0.2950	-0.5379	-4.24
2.255	0.2935	-0.5378	-4.26
2.260	0.2921	-0.5377	-4.27
2.265	0.2905	-0.5375	-4.28
2.270	0.2889	-0.5376	-4.29
2.275	0.2874	-0.5375	-4.30
2.280	0.2858	-0.5373	-4.31
2.285	0.2844	-0.5373	-4.32
2.290	0.2827	-0.5369	-4.34
2.295	0.2813	-0.5368	-4.35
2.300	0.2797	-0.5367	-4.36
2.305	0.2783	-0.5366	-4.37
2.310	0.2767	-0.5365	-4.38
2.315	0.2751	-0.5363	-4.40
2.320	0.2736	-0.5363	-4.41
2.325	0.2721	-0.5360	-4.42
2.330	0.2706	-0.5361	-4.43
2.335	0.2693	-0.5359	-4.44
2.340	0.2677	-0.5358	-4.45
2.345	0.2662	-0.5356	-4.46
2.350	0.2648	-0.5355	-4.48
2.355	0.2635	-0.5354	-4.48
2.360	0.2622	-0.5351	-4.50
2.365	0.2608	-0.5350	-4.51
2.370	0.2593	-0.5348	-4.52
2.375	0.2582	-0.5346	-4.53
2.380	0.2569	-0.5345	-4.54
2.385	0.2556	-0.5344	-4.55
2.390	0.2545	-0.5343	-4.56
2.395	0.2532	-0.5342	-4.57
2.400	0.2520	-0.5340	-4.58
2.405	0.2508	-0.5338	-4.59
2.410	0.2496	-0.5337	-4.59
2.415	0.2486	-0.5336	-4.60

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.2474	-0.5335	-4.61
2.425	0.2464	-0.5333	-4.62
2.430	0.2454	-0.5331	-4.63
2.435	0.2442	-0.5329	-4.64
2.440	0.2432	-0.5327	-4.65
2.445	0.2421	-0.5326	-4.66
2.450	0.2414	-0.5324	-4.66
2.455	0.2404	-0.5321	-4.67
2.460	0.2396	-0.5319	-4.68
2.465	0.2386	-0.5317	-4.69
2.470	0.2378	-0.5314	-4.70
2.475	0.2369	-0.5313	-4.71
2.480	0.2360	-0.5311	-4.71
2.485	0.2352	-0.5309	-4.72
2.490	0.2344	-0.5307	-4.73
2.495	0.2336	-0.5305	-4.74
2.500	0.2328	-0.5303	-4.74
2.505	0.2320	-0.5303	-4.75
2.510	0.2312	-0.5300	-4.76
2.515	0.2304	-0.5299	-4.76
2.520	0.2297	-0.5298	-4.77
2.525	0.2289	-0.5296	-4.78
2.530	0.2281	-0.5295	-4.78
2.535	0.2274	-0.5294	-4.79
2.540	0.2268	-0.5293	-4.79
2.545	0.2260	-0.5292	-4.80
2.550	0.2253	-0.5290	-4.81
2.555	0.2248	-0.5286	-4.82
2.560	0.2242	-0.5284	-4.82
2.565	0.2239	-0.5279	-4.83
2.570	0.2235	-0.5274	-4.84
2.575	0.2234	-0.5271	-4.84
2.580	0.2233	-0.5266	-4.85
2.585	0.2233	-0.5263	-4.86
2.590	0.2231	-0.5262	-4.86
2.595	0.2230	-0.5262	-4.86
2.600	0.2230	-0.5261	-4.86
2.605	0.2229	-0.5262	-4.86
2.610	0.2227	-0.5261	-4.86
2.615	0.2225	-0.5260	-4.87
2.620	0.2223	-0.5259	-4.87
2.625	0.2221	-0.5259	-4.87
2.630	0.2220	-0.5257	-4.87
2.635	0.2217	-0.5256	-4.88
2.640	0.2214	-0.5255	-4.88
2.645	0.2212	-0.5254	-4.88

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.2209	-0.5253	-4.89
2.655	0.2205	-0.5252	-4.89
2.660	0.2202	-0.5250	-4.89
2.665	0.2198	-0.5249	-4.90
2.670	0.2192	-0.5247	-4.90
2.675	0.2189	-0.5246	-4.91
2.680	0.2185	-0.5244	-4.91
2.685	0.2182	-0.5244	-4.91
2.690	0.2177	-0.5243	-4.92
2.695	0.2175	-0.5241	-4.92
2.700	0.2172	-0.5240	-4.93
2.705	0.2169	-0.5237	-4.93
2.710	0.2166	-0.5237	-4.93
2.715	0.2164	-0.5232	-4.94
2.720	0.2162	-0.5229	-4.95
2.725	0.2159	-0.5226	-4.95
2.730	0.2156	-0.5221	-4.96
2.735	0.2153	-0.5217	-4.97
2.740	0.2150	-0.5211	-4.98
2.745	0.2144	-0.5208	-4.99
2.750	0.2141	-0.5202	-5.00
2.755	0.2135	-0.5198	-5.01
2.760	0.2130	-0.5193	-5.02
2.765	0.2124	-0.5188	-5.03
2.770	0.2119	-0.5183	-5.04
2.775	0.2112	-0.5178	-5.05
2.780	0.2106	-0.5174	-5.06
2.785	0.2099	-0.5170	-5.07
2.790	0.2093	-0.5167	-5.08
2.795	0.2085	-0.5163	-5.09
2.800	0.2078	-0.5159	-5.10
2.805	0.2073	-0.5156	-5.10
2.810	0.2064	-0.5153	-5.11
2.815	0.2057	-0.5149	-5.12
2.820	0.2050	-0.5147	-5.13
2.825	0.2043	-0.5143	-5.14
2.830	0.2037	-0.5140	-5.15
2.835	0.2028	-0.5138	-5.15
2.840	0.2021	-0.5135	-5.16
2.845	0.2015	-0.5132	-5.17
2.850	0.2008	-0.5128	-5.18
2.855	0.2002	-0.5126	-5.19
2.860	0.1994	-0.5124	-5.20
2.865	0.1988	-0.5120	-5.20
2.870	0.1983	-0.5118	-5.21
2.875	0.1978	-0.5114	-5.22

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	0.1973	-0.5108	-5.23
2.885	0.1968	-0.5102	-5.24
2.890	0.1959	-0.5096	-5.26
2.895	0.1952	-0.5089	-5.27
2.900	0.1942	-0.5084	-5.28
2.905	0.1934	-0.5077	-5.30
2.910	0.1924	-0.5071	-5.31
2.915	0.1913	-0.5068	-5.33
2.920	0.1903	-0.5064	-5.34
2.925	0.1893	-0.5061	-5.35
2.930	0.1881	-0.5058	-5.36
2.935	0.1873	-0.5056	-5.37
2.940	0.1865	-0.5056	-5.37
2.945	0.1856	-0.5055	-5.38
2.950	0.1847	-0.5055	-5.38
2.955	0.1840	-0.5054	-5.39
2.960	0.1834	-0.5054	-5.39
2.965	0.1829	-0.5055	-5.39
2.970	0.1824	-0.5055	-5.39
2.975	0.1820	-0.5054	-5.40
2.980	0.1814	-0.5055	-5.40
2.985	0.1812	-0.5054	-5.40
2.990	0.1810	-0.5055	-5.40
2.995	0.1809	-0.5054	-5.40
3.000	0.1806	-0.5051	-5.41
3.005	0.1783	-0.5045	-5.43
3.010	0.1780	-0.5048	-5.43
3.015	0.1779	-0.5048	-5.43
3.020	0.1778	-0.5049	-5.43
3.025	0.1780	-0.5049	-5.43
3.030	0.1780	-0.5048	-5.43
3.035	0.1781	-0.5042	-5.44
3.040	0.1782	-0.5038	-5.44
3.045	0.1783	-0.5032	-5.45
3.050	0.1785	-0.5028	-5.46
3.055	0.1784	-0.5021	-5.47
3.060	0.1781	-0.5014	-5.48
3.065	0.1779	-0.5006	-5.49
3.070	0.1775	-0.4998	-5.51
3.075	0.1770	-0.4990	-5.52
3.080	0.1767	-0.4984	-5.53
3.085	0.1762	-0.4975	-5.55
3.090	0.1750	-0.4970	-5.57
3.095	0.1743	-0.4965	-5.58
3.100	0.1737	-0.4962	-5.59
3.105	0.1728	-0.4962	-5.59

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.1720	-0.4960	-5.60
3.115	0.1711	-0.4956	-5.61
3.120	0.1704	-0.4958	-5.61
3.125	0.1696	-0.4962	-5.61
3.130	0.1691	-0.4964	-5.61
3.135	0.1686	-0.4967	-5.60
3.140	0.1683	-0.4970	-5.60
3.145	0.1679	-0.4976	-5.59
3.150	0.1676	-0.4979	-5.59
3.155	0.1676	-0.4980	-5.59
3.160	0.1676	-0.4985	-5.58
3.165	0.1677	-0.4988	-5.58
3.170	0.1678	-0.4990	-5.57
3.175	0.1679	-0.4989	-5.57
3.180	0.1683	-0.4991	-5.57
3.185	0.1686	-0.4988	-5.57
3.190	0.1689	-0.4987	-5.57
3.195	0.1690	-0.4982	-5.58
3.200	0.1693	-0.4975	-5.59
3.205	0.1691	-0.4968	-5.60
3.210	0.1692	-0.4961	-5.61
3.215	0.1689	-0.4953	-5.62
3.220	0.1684	-0.4944	-5.64
3.225	0.1679	-0.4941	-5.65
3.230	0.1672	-0.4940	-5.66
3.235	0.1664	-0.4938	-5.66
3.240	0.1660	-0.4938	-5.66
3.245	0.1656	-0.4939	-5.66
3.250	0.1651	-0.4940	-5.67
3.255	0.1649	-0.4940	-5.67
3.260	0.1643	-0.4943	-5.66
3.265	0.1640	-0.4943	-5.67
3.270	0.1636	-0.4945	-5.67
3.275	0.1635	-0.4947	-5.66
3.280	0.1632	-0.4947	-5.66
3.285	0.1629	-0.4948	-5.66
3.290	0.1625	-0.4948	-5.67
3.295	0.1624	-0.4950	-5.66
3.300	0.1621	-0.4950	-5.67
3.305	0.1621	-0.4949	-5.67
3.310	0.1619	-0.4951	-5.67
3.315	0.1616	-0.4950	-5.67
3.320	0.1616	-0.4949	-5.67
3.325	0.1614	-0.4949	-5.67
3.330	0.1614	-0.4946	-5.68
3.335	0.1612	-0.4943	-5.68

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.1612	-0.4943	-5.68
3.345	0.1611	-0.4939	-5.69
3.350	0.1610	-0.4934	-5.70
3.355	0.1609	-0.4930	-5.70
3.360	0.1608	-0.4925	-5.71
3.365	0.1606	-0.4919	-5.72
3.370	0.1603	-0.4915	-5.73
3.375	0.1601	-0.4907	-5.74
3.380	0.1596	-0.4901	-5.76
3.385	0.1594	-0.4893	-5.77
3.390	0.1589	-0.4887	-5.78
3.395	0.1584	-0.4880	-5.80
3.400	0.1579	-0.4872	-5.81
3.405	0.1577	-0.4862	-5.83
3.410	0.1569	-0.4854	-5.85
3.415	0.1562	-0.4848	-5.86
3.420	0.1554	-0.4840	-5.88
3.425	0.1546	-0.4831	-5.90
3.430	0.1537	-0.4827	-5.91
3.435	0.1529	-0.4819	-5.92
3.440	0.1519	-0.4813	-5.94
3.445	0.1509	-0.4804	-5.96
3.450	0.1497	-0.4800	-5.97
3.455	0.1486	-0.4796	-5.98
3.460	0.1477	-0.4793	-5.99
3.465	0.1469	-0.4791	-6.00
3.470	0.1457	-0.4790	-6.01
3.475	0.1448	-0.4786	-6.02
3.480	0.1439	-0.4786	-6.02
3.485	0.1426	-0.4787	-6.03
3.490	0.1422	-0.4790	-6.03
3.495	0.1417	-0.4791	-6.03
3.500	0.1413	-0.4791	-6.03
3.505	0.1409	-0.4794	-6.03
3.510	0.1409	-0.4793	-6.03
3.515	0.1408	-0.4791	-6.03
3.520	0.1410	-0.4787	-6.04
3.525	0.1410	-0.4785	-6.04
3.530	0.1413	-0.4779	-6.05
3.535	0.1419	-0.4770	-6.06
3.540	0.1425	-0.4758	-6.08
3.545	0.1424	-0.4740	-6.11
3.550	0.1423	-0.4720	-6.14
3.555	0.1419	-0.4703	-6.17
3.560	0.1409	-0.4680	-6.22
3.565	0.1399	-0.4661	-6.26

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.570	0.1380	-0.4642	-6.30
3.575	0.1357	-0.4624	-6.34
3.580	0.1332	-0.4608	-6.38
3.585	0.1303	-0.4601	-6.41
3.590	0.1274	-0.4608	-6.41
3.595	0.1250	-0.4615	-6.41
3.600	0.1233	-0.4630	-6.39
3.605	0.1220	-0.4647	-6.37
3.610	0.1220	-0.4665	-6.34
3.615	0.1221	-0.4676	-6.32
3.620	0.1230	-0.4682	-6.30
3.625	0.1238	-0.4686	-6.29
3.630	0.1247	-0.4689	-6.28
3.635	0.1262	-0.4683	-6.29
3.640	0.1266	-0.4675	-6.30
3.645	0.1270	-0.4663	-6.32
3.650	0.1273	-0.4651	-6.34
3.655	0.1270	-0.4638	-6.36
3.660	0.1270	-0.4627	-6.38
3.665	0.1268	-0.4616	-6.40
3.670	0.1262	-0.4604	-6.42
3.675	0.1263	-0.4594	-6.44
3.680	0.1257	-0.4580	-6.47
3.685	0.1250	-0.4570	-6.49
3.690	0.1244	-0.4555	-6.52
3.695	0.1231	-0.4542	-6.55
3.700	0.1219	-0.4529	-6.58
3.705	0.1205	-0.4519	-6.60
3.710	0.1184	-0.4508	-6.63
3.715	0.1162	-0.4502	-6.65
3.720	0.1137	-0.4497	-6.67
3.725	0.1113	-0.4503	-6.67
3.730	0.1081	-0.4510	-6.67
3.735	0.1062	-0.4528	-6.65
3.740	0.1045	-0.4551	-6.62
3.745	0.1040	-0.4572	-6.58
3.750	0.1044	-0.4600	-6.53
3.755	0.1055	-0.4619	-6.49
3.760	0.1075	-0.4633	-6.45
3.765	0.1099	-0.4638	-6.44
3.770	0.1122	-0.4632	-6.44
3.775	0.1143	-0.4615	-6.46
3.780	0.1160	-0.4593	-6.49
3.785	0.1159	-0.4564	-6.54
3.790	0.1150	-0.4540	-6.59
3.795	0.1136	-0.4525	-6.62

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.1121	-0.4512	-6.65
3.805	0.1102	-0.4501	-6.68
3.810	0.1087	-0.4498	-6.69
3.815	0.1069	-0.4493	-6.71
3.820	0.1055	-0.4493	-6.72
3.825	0.1040	-0.4495	-6.72
3.830	0.1026	-0.4499	-6.72
3.835	0.1022	-0.4525	-6.67
3.840	0.1016	-0.4524	-6.68
3.845	0.1011	-0.4525	-6.68
3.850	0.1007	-0.4524	-6.68
3.855	0.1001	-0.4524	-6.68
3.860	0.0996	-0.4523	-6.69
3.865	0.0991	-0.4520	-6.69
3.870	0.0985	-0.4519	-6.70
3.875	0.0979	-0.4518	-6.70
3.880	0.0972	-0.4515	-6.71
3.885	0.0966	-0.4514	-6.71
3.890	0.0958	-0.4513	-6.72
3.895	0.0952	-0.4510	-6.73
3.900	0.0944	-0.4509	-6.73
3.905	0.0935	-0.4508	-6.74
3.910	0.0928	-0.4506	-6.74
3.915	0.0918	-0.4504	-6.75
3.920	0.0910	-0.4502	-6.76
3.925	0.0902	-0.4501	-6.76
3.930	0.0894	-0.4499	-6.77
3.935	0.0885	-0.4495	-6.78
3.940	0.0875	-0.4494	-6.79
3.945	0.0868	-0.4492	-6.79
3.950	0.0856	-0.4489	-6.80
3.955	0.0848	-0.4485	-6.81
3.960	0.0837	-0.4483	-6.82
3.965	0.0831	-0.4479	-6.83
3.970	0.0820	-0.4478	-6.84
3.975	0.0811	-0.4475	-6.84
3.980	0.0799	-0.4472	-6.85
3.985	0.0789	-0.4469	-6.86
3.990	0.0780	-0.4465	-6.87
3.995	0.0768	-0.4461	-6.88
4.000	0.0758	-0.4458	-6.89
4.005	0.0748	-0.4455	-6.90
4.010	0.0736	-0.4451	-6.91
4.015	0.0727	-0.4447	-6.92
4.020	0.0717	-0.4443	-6.94
4.025	0.0705	-0.4439	-6.95

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.030	0.0695	-0.4436	-6.95
4.035	0.0684	-0.4432	-6.97
4.040	0.0674	-0.4428	-6.98
4.045	0.0664	-0.4423	-6.99
4.050	0.0654	-0.4418	-7.00
4.055	0.0642	-0.4415	-7.01
4.060	0.0632	-0.4410	-7.02
4.065	0.0619	-0.4405	-7.04
4.070	0.0611	-0.4402	-7.05
4.075	0.0600	-0.4398	-7.05
4.080	0.0591	-0.4395	-7.06
4.085	0.0580	-0.4390	-7.08
4.090	0.0569	-0.4386	-7.09
4.095	0.0561	-0.4385	-7.09
4.100	0.0550	-0.4379	-7.10
4.105	0.0539	-0.4376	-7.11
4.110	0.0530	-0.4374	-7.12
4.115	0.0517	-0.4372	-7.13
4.120	0.0507	-0.4368	-7.14
4.125	0.0496	-0.4367	-7.14
4.130	0.0484	-0.4364	-7.15
4.135	0.0474	-0.4359	-7.16
4.140	0.0461	-0.4358	-7.17
4.145	0.0451	-0.4355	-7.17
4.150	0.0438	-0.4350	-7.19
4.155	0.0429	-0.4345	-7.20
4.160	0.0419	-0.4341	-7.21
4.165	0.0412	-0.4337	-7.22
4.170	0.0405	-0.4332	-7.23
4.175	0.0400	-0.4329	-7.24
4.180	0.0394	-0.4325	-7.24
4.185	0.0388	-0.4322	-7.25
4.190	0.0382	-0.4320	-7.26
4.195	0.0377	-0.4319	-7.26
4.200	0.0373	-0.4317	-7.26
4.205	0.0369	-0.4318	-7.26
4.210	0.0365	-0.4315	-7.27
4.215	0.0361	-0.4315	-7.27
4.220	0.0358	-0.4315	-7.27
4.225	0.0353	-0.4314	-7.27
4.230	0.0350	-0.4314	-7.27
4.235	0.0347	-0.4315	-7.27
4.240	0.0345	-0.4314	-7.27
4.245	0.0342	-0.4316	-7.27
4.250	0.0340	-0.4318	-7.27
4.255	0.0338	-0.4318	-7.27

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	0.0335	-0.4318	-7.27
4.265	0.0333	-0.4320	-7.26
4.270	0.0333	-0.4321	-7.26
4.275	0.0332	-0.4322	-7.26
4.280	0.0330	-0.4323	-7.26
4.285	0.0330	-0.4324	-7.26
4.290	0.0328	-0.4325	-7.26
4.295	0.0328	-0.4325	-7.26
4.300	0.0328	-0.4327	-7.25
4.305	0.0327	-0.4328	-7.25
4.310	0.0324	-0.4329	-7.25
4.315	0.0324	-0.4331	-7.24
4.320	0.0323	-0.4331	-7.24
4.325	0.0322	-0.4335	-7.24
4.330	0.0321	-0.4335	-7.24
4.335	0.0320	-0.4338	-7.23
4.340	0.0318	-0.4341	-7.23
4.345	0.0317	-0.4345	-7.22
4.350	0.0318	-0.4349	-7.21
4.355	0.0318	-0.4352	-7.20
4.360	0.0320	-0.4356	-7.19
4.365	0.0321	-0.4362	-7.18
4.370	0.0322	-0.4365	-7.18
4.375	0.0326	-0.4369	-7.17
4.380	0.0329	-0.4373	-7.16
4.385	0.0330	-0.4376	-7.15
4.390	0.0336	-0.4380	-7.14
4.395	0.0340	-0.4384	-7.14
4.400	0.0344	-0.4386	-7.13
4.405	0.0348	-0.4388	-7.13
4.410	0.0353	-0.4389	-7.12
4.415	0.0357	-0.4390	-7.12
4.420	0.0359	-0.4392	-7.12
4.425	0.0363	-0.4390	-7.12
4.430	0.0367	-0.4391	-7.12
4.435	0.0367	-0.4390	-7.12
4.440	0.0369	-0.4390	-7.12
4.445	0.0372	-0.4390	-7.12
4.450	0.0371	-0.4388	-7.12
4.455	0.0373	-0.4390	-7.12
4.460	0.0377	-0.4389	-7.12
4.465	0.0379	-0.4392	-7.12
4.470	0.0385	-0.4394	-7.11
4.475	0.0389	-0.4399	-7.10
4.480	0.0395	-0.4404	-7.09
4.485	0.0398	-0.4410	-7.08

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	0.0404	-0.4416	-7.06
4.495	0.0409	-0.4422	-7.05
4.500	0.0416	-0.4429	-7.04
4.505	0.0420	-0.4434	-7.02
4.510	0.0427	-0.4438	-7.02
4.515	0.0433	-0.4439	-7.01
4.520	0.0440	-0.4440	-7.01
4.525	0.0445	-0.4439	-7.01
4.530	0.0449	-0.4439	-7.01
4.535	0.0452	-0.4437	-7.01
4.540	0.0455	-0.4432	-7.02
4.545	0.0457	-0.4430	-7.03
4.550	0.0458	-0.4427	-7.03
4.555	0.0458	-0.4426	-7.03
4.560	0.0459	-0.4424	-7.04
4.565	0.0460	-0.4422	-7.04
4.570	0.0460	-0.4419	-7.05
4.575	0.0460	-0.4413	-7.06
4.580	0.0460	-0.4411	-7.06
4.585	0.0462	-0.4409	-7.07
4.590	0.0461	-0.4404	-7.08
4.595	0.0460	-0.4401	-7.08
4.600	0.0457	-0.4397	-7.09
4.605	0.0455	-0.4392	-7.10
4.610	0.0454	-0.4387	-7.11
4.615	0.0451	-0.4384	-7.12
4.620	0.0452	-0.4380	-7.12
4.625	0.0449	-0.4378	-7.13
4.630	0.0445	-0.4375	-7.14
4.635	0.0444	-0.4370	-7.15
4.640	0.0443	-0.4367	-7.15
4.645	0.0443	-0.4365	-7.16
4.650	0.0441	-0.4361	-7.16
4.655	0.0438	-0.4355	-7.18
4.660	0.0441	-0.4354	-7.18
4.665	0.0441	-0.4350	-7.19
4.670	0.0439	-0.4349	-7.19
4.675	0.0442	-0.4345	-7.20
4.680	0.0441	-0.4342	-7.20
4.685	0.0443	-0.4338	-7.21
4.690	0.0442	-0.4334	-7.22
4.695	0.0443	-0.4333	-7.22
4.700	0.0444	-0.4327	-7.23
4.705	0.0445	-0.4321	-7.24
4.710	0.0443	-0.4318	-7.25
4.715	0.0442	-0.4312	-7.26

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.720	0.0440	-0.4306	-7.27
4.725	0.0437	-0.4302	-7.28
4.730	0.0436	-0.4296	-7.29
4.735	0.0431	-0.4287	-7.31
4.740	0.0425	-0.4281	-7.33
4.745	0.0419	-0.4274	-7.34
4.750	0.0412	-0.4269	-7.35
4.755	0.0405	-0.4263	-7.37
4.760	0.0399	-0.4259	-7.38
4.765	0.0392	-0.4256	-7.38
4.770	0.0388	-0.4252	-7.39
4.775	0.0384	-0.4250	-7.40
4.780	0.0382	-0.4246	-7.41
4.785	0.0380	-0.4246	-7.41
4.790	0.0379	-0.4240	-7.42
4.795	0.0381	-0.4235	-7.43
4.800	0.0382	-0.4231	-7.44
4.805	0.0384	-0.4228	-7.44
4.810	0.0385	-0.4221	-7.46
4.815	0.0388	-0.4214	-7.47
4.820	0.0390	-0.4208	-7.48
4.825	0.0394	-0.4202	-7.49
4.830	0.0393	-0.4198	-7.50
4.835	0.0394	-0.4194	-7.51
4.840	0.0395	-0.4190	-7.52
4.845	0.0395	-0.4188	-7.52
4.850	0.0396	-0.4184	-7.53
4.855	0.0396	-0.4185	-7.53
4.860	0.0394	-0.4183	-7.53
4.865	0.0394	-0.4182	-7.53
4.870	0.0391	-0.4183	-7.53
4.875	0.0389	-0.4184	-7.53
4.880	0.0387	-0.4185	-7.53
4.885	0.0389	-0.4188	-7.52
4.890	0.0384	-0.4190	-7.52
4.895	0.0382	-0.4191	-7.52
4.900	0.0381	-0.4193	-7.51
4.905	0.0379	-0.4194	-7.51
4.910	0.0377	-0.4196	-7.51
4.915	0.0376	-0.4198	-7.50
4.920	0.0374	-0.4199	-7.50
4.925	0.0373	-0.4201	-7.50
4.930	0.0371	-0.4203	-7.50
4.935	0.0370	-0.4205	-7.49
4.940	0.0368	-0.4206	-7.49
4.945	0.0366	-0.4207	-7.49

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.0366	-0.4209	-7.48
4.955	0.0364	-0.4210	-7.48
4.960	0.0362	-0.4211	-7.48
4.965	0.0362	-0.4214	-7.47
4.970	0.0361	-0.4215	-7.47
4.975	0.0360	-0.4217	-7.47
4.980	0.0359	-0.4217	-7.47
4.985	0.0358	-0.4218	-7.47
4.990	0.0358	-0.4219	-7.47
4.995	0.0358	-0.4222	-7.46
5.000	0.0358	-0.4222	-7.46
5.005	0.0360	-0.4225	-7.45
5.010	0.0359	-0.4226	-7.45
5.015	0.0361	-0.4229	-7.44
5.020	0.0361	-0.4229	-7.44
5.025	0.0363	-0.4232	-7.44
5.030	0.0364	-0.4232	-7.44
5.035	0.0365	-0.4235	-7.43
5.040	0.0367	-0.4233	-7.43
5.045	0.0370	-0.4236	-7.43
5.050	0.0373	-0.4235	-7.43
5.055	0.0374	-0.4237	-7.43
5.060	0.0376	-0.4234	-7.43
5.065	0.0377	-0.4237	-7.42
5.070	0.0380	-0.4234	-7.43
5.075	0.0378	-0.4232	-7.43
5.080	0.0381	-0.4230	-7.44
5.085	0.0378	-0.4227	-7.44
5.090	0.0381	-0.4225	-7.45
5.095	0.0378	-0.4222	-7.46
5.100	0.0380	-0.4219	-7.46
5.105	0.0377	-0.4215	-7.47
5.110	0.0377	-0.4213	-7.47
5.115	0.0376	-0.4210	-7.48
5.120	0.0377	-0.4207	-7.49
5.125	0.0374	-0.4204	-7.49
5.130	0.0375	-0.4202	-7.50
5.135	0.0374	-0.4199	-7.50
5.140	0.0373	-0.4196	-7.51
5.145	0.0374	-0.4193	-7.52
5.150	0.0371	-0.4190	-7.52
5.155	0.0369	-0.4185	-7.53
5.160	0.0367	-0.4182	-7.54
5.165	0.0364	-0.4177	-7.55
5.170	0.0362	-0.4172	-7.56
5.175	0.0359	-0.4168	-7.57

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.0355	-0.4162	-7.58
5.185	0.0350	-0.4156	-7.60
5.190	0.0347	-0.4150	-7.61
5.195	0.0342	-0.4144	-7.62
5.200	0.0337	-0.4138	-7.64
5.205	0.0331	-0.4130	-7.65
5.210	0.0328	-0.4125	-7.66
5.215	0.0323	-0.4118	-7.68
5.220	0.0318	-0.4111	-7.70
5.225	0.0314	-0.4102	-7.72
5.230	0.0307	-0.4095	-7.73
5.235	0.0302	-0.4087	-7.75
5.240	0.0296	-0.4079	-7.77
5.245	0.0289	-0.4071	-7.78
5.250	0.0284	-0.4062	-7.80
5.255	0.0278	-0.4052	-7.83
5.260	0.0270	-0.4045	-7.84
5.265	0.0263	-0.4035	-7.86
5.270	0.0255	-0.4029	-7.88
5.275	0.0249	-0.4021	-7.90
5.280	0.0243	-0.4011	-7.92
5.285	0.0234	-0.4005	-7.93
5.290	0.0229	-0.3999	-7.95
5.295	0.0222	-0.3992	-7.96
5.300	0.0214	-0.3985	-7.98
5.305	0.0208	-0.3981	-7.99
5.310	0.0200	-0.3972	-8.01
5.315	0.0193	-0.3967	-8.02
5.320	0.0188	-0.3960	-8.04
5.325	0.0181	-0.3953	-8.05
5.330	0.0175	-0.3947	-8.07
5.335	0.0170	-0.3938	-8.09
5.340	0.0163	-0.3932	-8.10
5.345	0.0157	-0.3925	-8.12
5.350	0.0151	-0.3919	-8.13
5.355	0.0145	-0.3910	-8.15
5.360	0.0139	-0.3903	-8.17
5.365	0.0132	-0.3895	-8.19
5.370	0.0125	-0.3887	-8.20
5.375	0.0118	-0.3878	-8.22
5.380	0.0111	-0.3869	-8.24
5.385	0.0101	-0.3861	-8.26
5.390	0.0092	-0.3855	-8.28
5.395	0.0083	-0.3848	-8.29
5.400	0.0075	-0.3842	-8.31
5.405	0.0064	-0.3839	-8.31

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.0056	-0.3833	-8.33
5.415	0.0046	-0.3828	-8.34
5.420	0.0040	-0.3823	-8.35
5.425	0.0033	-0.3819	-8.36
5.430	0.0023	-0.3813	-8.37
5.435	0.0015	-0.3811	-8.38
5.440	0.0008	-0.3807	-8.39
5.445	0.0000	-0.3802	-8.40
5.450	-0.0007	-0.3800	-8.40
5.455	-0.0012	-0.3798	-8.41
5.460	-0.0019	-0.3793	-8.42
5.465	-0.0024	-0.3789	-8.43
5.470	-0.0032	-0.3784	-8.44
5.475	-0.0039	-0.3779	-8.45
5.480	-0.0045	-0.3775	-8.46
5.485	-0.0053	-0.3770	-8.47
5.490	-0.0061	-0.3767	-8.48
5.495	-0.0069	-0.3762	-8.49
5.500	-0.0078	-0.3759	-8.50
5.505	-0.0086	-0.3756	-8.50
5.510	-0.0096	-0.3755	-8.50
5.515	-0.0104	-0.3755	-8.51
5.520	-0.0112	-0.3754	-8.51
5.525	-0.0117	-0.3755	-8.50
5.530	-0.0125	-0.3756	-8.50
5.535	-0.0130	-0.3757	-8.50
5.540	-0.0137	-0.3758	-8.50
5.545	-0.0142	-0.3759	-8.49
5.550	-0.0148	-0.3759	-8.49
5.555	-0.0152	-0.3760	-8.49
5.560	-0.0158	-0.3761	-8.49
5.565	-0.0164	-0.3762	-8.48
5.570	-0.0169	-0.3761	-8.48
5.575	-0.0176	-0.3762	-8.48
5.580	-0.0182	-0.3762	-8.48
5.585	-0.0189	-0.3762	-8.48
5.590	-0.0195	-0.3764	-8.48
5.595	-0.0203	-0.3765	-8.47
5.600	-0.0210	-0.3767	-8.47
5.605	-0.0218	-0.3770	-8.46
5.610	-0.0225	-0.3774	-8.45
5.615	-0.0233	-0.3779	-8.44
5.620	-0.0239	-0.3782	-8.43
5.625	-0.0244	-0.3788	-8.41
5.630	-0.0249	-0.3792	-8.40
5.635	-0.0253	-0.3797	-8.39

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.0257	-0.3801	-8.38
5.645	-0.0261	-0.3806	-8.37
5.650	-0.0263	-0.3810	-8.36
5.655	-0.0267	-0.3813	-8.35
5.660	-0.0270	-0.3816	-8.35
5.665	-0.0274	-0.3818	-8.34
5.670	-0.0277	-0.3822	-8.33
5.675	-0.0280	-0.3826	-8.32
5.680	-0.0285	-0.3829	-8.31
5.685	-0.0288	-0.3832	-8.31
5.690	-0.0291	-0.3834	-8.30
5.695	-0.0292	-0.3838	-8.29
5.700	-0.0294	-0.3840	-8.29
5.705	-0.0296	-0.3842	-8.28
5.710	-0.0298	-0.3845	-8.28
5.715	-0.0298	-0.3847	-8.27
5.720	-0.0300	-0.3849	-8.27
5.725	-0.0301	-0.3851	-8.26
5.730	-0.0303	-0.3855	-8.25
5.735	-0.0304	-0.3855	-8.25
5.740	-0.0304	-0.3856	-8.25
5.745	-0.0305	-0.3857	-8.25
5.750	-0.0305	-0.3859	-8.24
5.755	-0.0306	-0.3859	-8.24
5.760	-0.0307	-0.3859	-8.24
5.765	-0.0308	-0.3861	-8.24
5.770	-0.0309	-0.3859	-8.24
5.775	-0.0309	-0.3856	-8.25
5.780	-0.0310	-0.3854	-8.25
5.785	-0.0311	-0.3854	-8.25
5.790	-0.0312	-0.3851	-8.26
5.795	-0.0314	-0.3849	-8.26
5.800	-0.0314	-0.3846	-8.27
5.805	-0.0314	-0.3841	-8.28
5.810	-0.0314	-0.3838	-8.29
5.815	-0.0315	-0.3835	-8.30
5.820	-0.0318	-0.3829	-8.31
5.825	-0.0320	-0.3827	-8.31
5.830	-0.0321	-0.3820	-8.33
5.835	-0.0322	-0.3814	-8.34
5.840	-0.0324	-0.3810	-8.35
5.845	-0.0329	-0.3804	-8.36
5.850	-0.0331	-0.3800	-8.37
5.855	-0.0333	-0.3792	-8.39
5.860	-0.0337	-0.3787	-8.40
5.865	-0.0338	-0.3780	-8.42

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.0344	-0.3775	-8.43
5.875	-0.0348	-0.3769	-8.44
5.880	-0.0352	-0.3764	-8.45
5.885	-0.0356	-0.3757	-8.46
5.890	-0.0362	-0.3753	-8.47
5.895	-0.0364	-0.3751	-8.48
5.900	-0.0368	-0.3744	-8.49
5.905	-0.0372	-0.3739	-8.50
5.910	-0.0377	-0.3737	-8.51
5.915	-0.0379	-0.3732	-8.52
5.920	-0.0382	-0.3728	-8.53
5.925	-0.0384	-0.3722	-8.54
5.930	-0.0388	-0.3717	-8.55
5.935	-0.0392	-0.3710	-8.56
5.940	-0.0394	-0.3706	-8.57
5.945	-0.0399	-0.3699	-8.59
5.950	-0.0403	-0.3692	-8.60
5.955	-0.0408	-0.3684	-8.62
5.960	-0.0412	-0.3679	-8.63
5.965	-0.0419	-0.3672	-8.65
5.970	-0.0428	-0.3666	-8.66
5.975	-0.0435	-0.3664	-8.66
5.980	-0.0443	-0.3660	-8.67
5.985	-0.0452	-0.3658	-8.67
5.990	-0.0457	-0.3656	-8.67
5.995	-0.0466	-0.3656	-8.67
6.000	-0.0470	-0.3656	-8.67

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.